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Bachelor of Science in Physics

JOHN SEN B. PRUDENTE

SHORT-TERM ORBITAL DYNAMICS OF HYPOTHETICAL HABITABLE ZONE TROJANS IN MULTI-PLANETARY SYSTEMS

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DECEMBER 2024

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JOHN SEN B. PRUDENTE

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SHORT-TERM ORBITAL DYNAMICS OF HYPOTHETICAL HABITABLE ZONE TROJANS IN MULTI-PLANETARY SYSTEMS

John Sen B. Prudente

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In Partial Fulfillment of the Requirements
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December 2024

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ANNOUNCEMENT OF THE BACHELOR'S EXAMINATION

of

JOHN SEN B. PRUDENTE

in defense of his Bachelor's Thesis

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for the degree of B.S. Physics

8:00 - 9:15 A.M., 9 DECEMBER 2024
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Dear Dean Peralta,,

Below is the official action of the Thesis Reader and the Panel Members in the undergraduate thesis defense of John Sen B. Prudente. The defense of the thesis, entitled **"SHORT-TERM ORBITAL DYNAMICS OF HYPOTHETICAL HABITABLE ZONE TROJANS IN MULTI-PLANETARY SYSTEMS,"** was held at Abraham Sarmiento Hall, Monday,09 December 2024, 8:00 AM - 9:30 AM.

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CERTIFICATION

This is to certify that this bachelor's thesis entitled, SHORT-TERM ORBITAL DYNAMICS OF HYPOTHETICAL HABITABLE ZONE TROJANS IN MULTI-PLANETARY SYSTEMS and submitted by JOHN SEN B. PRUDENTE to fulfill part of the requirements for the degree of Bachelor of Science in Physics was successfully defended and approved on 09 December 2024.

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ABSTRACT

SHORT-TERM ORBITAL DYNAMICS OF HYPOTHETICAL HABITABLE ZONE TROJANS IN MULTI-PLANETARY SYSTEMS

John Sen B. Prudente **Adviser: Crismar Patacsil, Ph. D.**
University of the Philippines - Baguio, 2024 **Co-Adviser: Christian Crisostomo, Ph. D.**

Finding habitable exoplanets is an expanding field of research in astronomy, and as such, understanding what exoplanetary system architecture plays a key role in finding potentially habitable worlds. This study will investigate Trojan bodies, astronomical objects co-orbiting a larger planet in the L_4 and L_5 Lagrange points, and their short-term stability around gas giants in the habitable zones of multi-planetary systems. The WHFast integrator in the REBOUND software was used to simulate the orbital dynamics of Trojans in 12,500 distinct N-body systems $(N = 4, 5)$ of two selected exoplanets - 47 UMa and HD 108874. Each simulation included an Earth-mass body positioned randomly near the L_4 Lagrangian points of its host planet and was integrated for 1000 orbital periods of the host gas giant. Orbital parameter plots and MEGNO stability criteria analyses, combined with comparison from previous studies, suggest that the stability of these hypothetical Trojans is influenced by the gas giant's mass, number of planets, and the initial orbital configuration of the Trojan body. This research highlights the value of including Trojan-hosting multi-planetary systems in future searches for habitable worlds.

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SHORT-TERM ORBITAL DYNAMICS OF HYPOTHETICAL HABITABLE ZONE TROJANS IN MULTI-PLANETARY SYSTEMS

Chapter 1

INTRODUCTION

1 Introduction

1.1 Purpose and Motivation

Since the first exoplanet was discovered around a pulsar in early 1992 by Wolszczan and Frail [1], exoplanet study and the search for life on other worlds has only evolved and grown in the past decades. Until recently, more than 5000 exoplanet discoveries have revealed a diverse array of planetary systems [2, 3, 4], placing astronomy in a time of population-level statistics for exoplanets [5]. With the launch of the James Webb Space Telescope (JWST), a shift in focus in finding potentially habitable worlds has been made. The diversity of planetary configurations in the universe underscores the need to understand different system architectures, especially in the context of planetary stability. As such, the importance of studying exoplanetary systems and their architecture continues to be an important part of astronomical studies, which includes studies of other astronomical objects such as Trojans and exomoons.

One primary planetary type common in exoplanetary systems are gas giants, a category of planets consisting predominantly of hydrogen and helium gases [6]. Gas giants' large masses significantly influence the orbits of other objects, shaping the architecture of exoplanetary systems. Their large masses play a crucial role in exoplanetary system dynamics, influencing the stability of objects in their vicinity due to their significant gravitational effects. These include exomoons and Trojan bodies. Studies have proposed that several discovered exoplanetary systems that host gas giants could hold potentially stable Trojan bodies in their orbit [7, 8].

Given their immense mass and gravitational influence, gas giants play a pivotal role in shaping the orbital architecture and stability of planetary systems. Their presence determines whether neighboring bodies maintain long-term, predictable trajectories or succumb to chaotic interactions [6]. Planetary stability is a critical focus in exoplanetary research and broader physics due to its intrinsic connection to the N-body problem—a complex, unsolved problem lacking analytic solutions. [9] Consequently, approximations of stable orbits are essential for understanding the longevity and predictability of planetary configurations. As the number of planets in a system increases, the likelihood of chaotic motion grows, making stability more challenging to achieve. Factors such as the Hill radius (the distance before a planet's gravitational effects recede), planetary composition, and orbital parameters such as eccentricities and inclinations of each planet further influence the stability of the entire system. [10, 11]

The nature of three-body systems provides unique solutions that depend on a multitude of factors. One significant example of such solutions are Trojans—celestial bodies of small mass sharing the orbit of a larger orbiting body, such as a planet. Trojans are found at one of the stable Lagrange points, specifically the L_4 and L_5 points, which are positions in orbital dynamics where gravitational forces and the motion of the smaller body balance out, allowing them to remain relatively stable (the exact stability mechanisms are explored in the literature section). These points emerge as solutions to the restricted three-body problem, where the system consists of a dominant central mass, a smaller orbiting body, and a third, significantly less massive body.

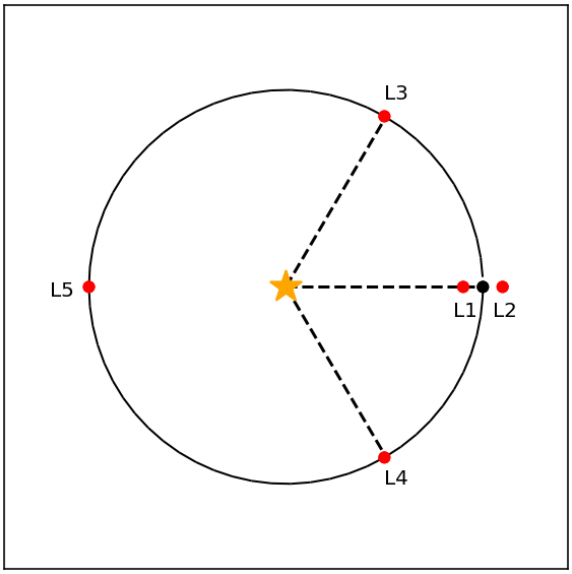

Figure 1: Lagrange points of the restricted three-body system, labeled L_1 to L_5.

Trojan bodies are of particular interest in planetary science because their stable orbits can provide valuable insights into the formation and evolution of planetary systems, both in our Solar System and in exoplanetary contexts [12]. In the Solar System, numerous Trojan asteroids are observed at the Jovian and Neptunian orbits, forming distinct "camps" ahead of and behind these gas giants. In the context of this study however, they will serve as hypothetical planets of potential habitability.

Although only Trojan asteroids are currently observed in the Solar System, the concept of terrestrial Trojan planets, where instead of asteroid a larger mass body such as a planet is in the Lagrange point, in exoplanetary systems has been hypothesized. Such configurations are believed to be highly stable and potentially observable [13]. Recent observations by the James Webb Space Telescope (JWST) have even identified a potential developing Trojan body in a forming planetary system [14]. This discovery, part of the TROY project, aims to uncover evidence for exotrojan bodies, offering intriguing

possibilities for planetary architectures involving planetary-mass Trojans.

Schwarz's work [8] has been instrumental in exploring the orbital dynamics of hypothetical Trojan bodies co-orbiting gas giants within the habitable zones of single-star systems. Their research has identified four distinct configurations of gas giants and terrestrial planets in habitable zones, shedding light on the potential stability of such systems and contributing valuable insights into exoplanetary dynamics.

Table 1: Possible planetary system configurations of a gas giant and a terrestrial planet in the habitable zone and their impact on habitability.

Label	**Description**
OHZ	The habitable zone is outside the orbit of a gas giant orbiting extremely close to the host star. However, if the gas giant's gravitational effects reach the habitable zone, gravitational perturbation can push the terrestrial planet out of the habitable zone.
SOL	When the gas giant moves far away from the central star, it allows additional planets to move on stable orbits closer to the star inside the habitable zone. This is best seen in the Solar System, where Jupiter (and other gas giants) are farther from the terrestrial inner planets.
SAT	When the gas giant itself moves in the habitable zone, a habitable natural satellite could have a stable orbit around the gas giant. Examples include the moons Titan and Europa in the Solar System.
TROJ	A Trojan-like habitable planet may move in a stable orbit around the Lagrangian equilibrium points L_4 or L_5 of a gas giant orbiting in the habitable zone.

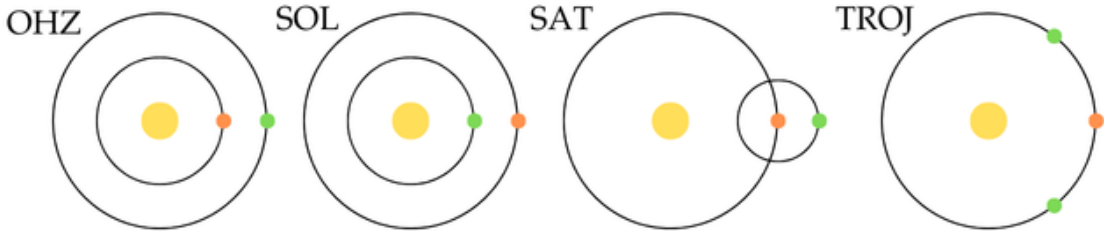

Figure 2: Illustration of possible configurations of a gas giant and a terrestrial planet and their impact on habitability.

Figure 2 shows an illustration of the different configurations defined by Schwarz' and previous work, where the the central yellow object is the star, the orange body is the gas giant, and the green body is the terrestrial planet. Following from Schwarz' and other related studies [15], this study will focus on the fourth configuration. Orbital stability studies of Trojans in gas giant orbits is used to understand complex exoplanetary system architectures. However, despite their study providing key insights into stable Trojans in gas giant orbits, one aspect of their work is to use exoplanetary system with one planet for most chosen exoplanetary system, which implies that it does not fully address how these dynamics might change in multi-planet systems. This presents a gap in the understanding of Trojan stability in exoplanetary system, which are a significant fraction of observed stellar systems in the universe.

Multi-planet systems represent a more complex and less explored environment for studying Trojan stability, which could yield different results due to the dual gravitational influence. The gravitational influence of other planets, depending on their properties, introduces additional complexities, such as periodic variations in orbital dynamics and potential destabilizing forces, which have not been thoroughly studied in the context of Trojan stability [16].

While the stability regions of Trojan bodies have been the work of many astrodynamics studies before [7, 8, 17], the effects of neighboring planets on their orbital stability have not been comprehensively studied. This gap in research restricts the understanding of how external gravitational influences impact the potential for short-term stability of these bodies in habitable regions. Furthermore, they assume negligible mass particles for Trojans. This study aims to improve their work by adding additional constraints, as well as using updated constants for determining habitable zone zones. Additionally, the list of confirmed exoplanets has increased since then, and as such their method needs to be applied to the updated database.

The primary objective of this study is to utilize numerical simulations via the REBOUND module software as well as the WHFast integrator to determine what factors are necessary in short-term stability of Trojans co-orbiting gas giants in the habitable zone of multi-planet systems, focusing on their orbital evolution over time. This work will provide a quantitative analysis of short-term stabilities of Trojans in multi-planetary exoplanetary system using stability the criteria MEGNO, as well as a comparative analysis between initial and final parameters. The research will create a comparative assessment of previous studies to the experimental findings. Furthermore, by simulating a range of multi-planet system configurations, the study will assess the effects of planetary multiplicity on these systems' dynamics. Lastly, this paper aims to contribute detailed simulations and results that add to the knowledge of Trojan systems in exoplanetary research.

1.2 Scope and Limitations

This study examines the short-term stability of Trojan bodies orbiting gas giants in multi-planet systems, particularly within the habitable zone, using planetary parameters from the NASA Exoplanetary Archive, which, as of October 2024, lists 5766 confirmed exoplanets, offering a rich source of potential candidates for simulation.

The simulations were done using the REBOUND software, and the chosen numerical integrator is WHFast, a symplectic integrator based on the Wisdom-Holman method. Integrator limitations, such as close-encounters and prolonged error accumulations apply (further explored in the relevant literature).

This study does not cover other aspects of orbital dynamics and habitability, and as such are subject to limitations such as:

Use of Initial Parameters. For the purposes of this experiment, simulations will use varying semi-major axis and true anomaly, and therefore lack the effects of varying other orbital parameters such as inclination or eccentricity. This study also only considers L_4 Trojans and does not imply similar results to L_5 Trojans due to gravitational effects of other planets in the system. Additionally, there is a manual addition of the planets in the same planetary system after the filtering process, which could introduce a bias in the data used for the simulations. The timescale of the system will also run only for 1000 orbital periods, showcasing short-term dynamics. Related studies provide simulation timescales by up to 10^4 to 10^7 orbital periods, which provide more realistic approximations. These limitations are primarily due to the computational power available.

Neglect of Other Physical Effects. For simplicity and computational efficiency, other forces such as tidal, relativistic corrections, or magnetic interactions are neglected due to the assumption of point-mass objects by the software. These forces, while significant in certain contexts, are assumed to have minimal impact on the primary orbital dynamics under investigation (compared to related studies [8, 17].

Possibility of Habitability. This study focuses exclusively on the orbital stability of hypothetical Trojan bodies around gas giants. While being inside of the star's habitable zone is a key aspect of a planet's capability for life, determining true habitability involves additional factors, such as atmospheric conditions, insolation collection, and internal planetary mechanisms [18], which are outside the scope of purely dynamical stability simulations.

Model Assumptions. The simulations do not account for events that could alter dynamics of the objects. Such assumptions include neglecting stellar evolution or motion and ignoring external perturbations such as passing planetary objects (asteroids, comets, etc). This is due to the limited capabilities of the software and the integrator.

Accuracy of Numerical Simulations. WHFast is designed for speed and efficiency but comes with limitations in accuracy. For highly chaotic systems or over extended timescales, slight numerical errors may accumulate, potentially affecting the precision of the short-term stability results. This is a limitation of all integrators as they approximate the numerical solutions of the equations of motion, and as such the use of other integrators are necessary as validation of this research. An error analysis test is done i the appendix.

The following sections provide a detailed overview of this study: Chapter 2 discusses the theoretical background for multi-planet systems, the numerical solvers, as well as stability criteria, as well as relevant literature, Chapter 3 describes the methodology, including the simulation setup using REBOUND, storing of values, and plotting of heatmaps; Chapter 4 presents the results and analysis of the short-term stability of Trojan bodies; and Chapter 5 concludes with the implications of these findings and future research directions.

Chapter 2

REVIEW OF RELATED LITERATURE

2 Review of Related Literature

2.1 Trojans and the Restricted Three-Body Problem

Trojan bodies represent a unique solution to the 'restricted' classical three-body problem, first predicted by Joseph-Louis Lagrange [19], shown in Figure 3. Due to the chaotic nature of the problem, Lagrange had to use approximations in terms of the masses, which is what is referred to by the term "restricted". Here, $M_1 \gg M_2 \gg m$.

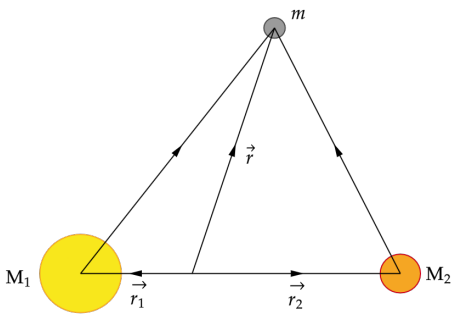

Figure 3: The restricted three-body problem.

The procedure for solving the problem is by seeking solutions to the equations of motion which maintain a constant separation between the three bodies, given as:

$$\vec{F} = -\frac{GM_1 m}{|\vec{r} - \vec{r_1}|^3}(\vec{r} - \vec{r_1}) - \frac{GM_2 m}{|\vec{r} - \vec{r_2}|^3}(\vec{r} - \vec{r_2}) \tag{1}$$

where both $\vec{r_1}$ and $\vec{r_1}$ are functions of time since M_1 and M_2 are orbiting each other. To simplify analysis, the problem is studied in a rotating reference frame, keeping the relative positions of the three bodies fixed, and the stationary solutions are what are defines the Lagrange points. In the rotating frame, the forces acting on the small body are the gravitational forces from M_1 and M_2 and the centrifugal force due to the rotating

reference frame. Equilibrium occurs where the gradient of this potential (visualized in Figure 4 is zero, or where the combined gravitational pull from the two large bodies plus the centrifugal force balance out.

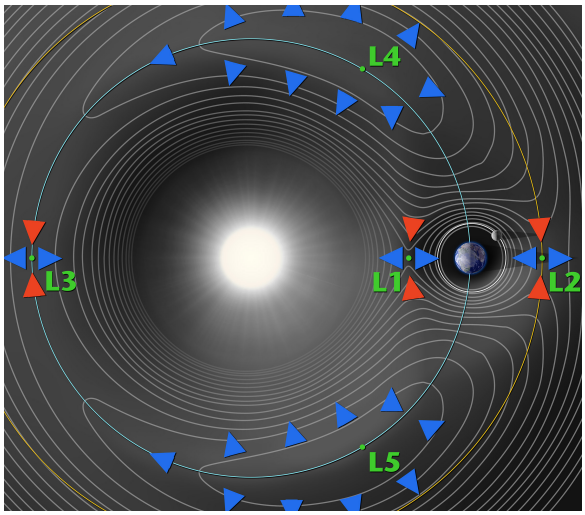

Figure 4: Diagram of the gravitational potential (white lines) associated with the Sun-Earth system. Credit: NASA / WMAP Science Team [20].

The generalized potential simplifies the analysis by encapsulating all forces into a single scalar field, enabling straightforward identification of equilibrium points and their stability while providing a clear visualization of the system's dynamics. In this potential field, there are 5 extrema in total: L_1, L_2, and L_3 (along the line connecting the two masses, unstable) and L_4 and L_5 (forming equilateral triangles with the masses, stable under certain conditions). The other Lagrange points are used by satellites for different purposes (such as JWST orbiting around (not inside) the L_2 point.)

In general, the Lagrange points of a two-body system, like the Earth-Sun or Earth-Moon system, are mainly influenced by the gravitational forces of the two primary bodies. However, the presence of other planets can still have an indirect effect on these points, particularly in multi-body systems. While other planets do not directly affect

the location of Lagrange points in a two-body system, they can introduce perturbations. For example, Saturn's gravitational influence can perturb the orbits of objects in the Jupiter-Sun Lagrange points, potentially destabilizing or altering their paths over long timescales. This can lead to changes in the dynamics of objects located at these points, such as asteroids or spacecraft. [21]

Such examples of Trojans include 2010 TK7, an Earth Trojan asteroid residing in the L_4 point of the Sun-Earth system [22], or the numerous Jupiter and Neptune Trojan asteroids distributed in so called "camps" [23]. All Trojans exhibit 1:1 mean motion resonance with their host planets. Although Trojans co-orbits a main planet, their orbital path is not always identical. Trojans are known to orbit in so-called "horseshoe" or "tadpole" orbital paths, where the Trojans oscillates around the L_4 or L_5 points.

Trojans are formed from within the protoplanetary disk that creates a planetary system. Although asteroids are their common form in our Solar System, Trojan planets have been proven to be observable in nature. A study by Cresswell and Nelson (2009) [13] examines the stability of low-mass Trojan planets that form in a protoplanetary disc and later accrete gas to become gas giants. Using two-dimensional hydrodynamics combined with N-body simulations, the authors model the evolution of Trojan pairs during and after gas disc dispersal, considering various initial masses, gas accretion rates, and disc properties. Results show that these systems remain stable throughout their evolution, with eccentricities staying low ($e < 0.02$) and stability persisting for at least 10^9 years after disc dispersal. Although large-scale inward migration can increase libration amplitudes and potentially disrupt Trojans before disc dispersal, the systems

are otherwise robust during transitions between different migration regimes and disc phases.

Schwarz's extensive work on stability regions of hypothetical Trojan bodies serves as the foundation of this study [8]. In their paper "Survey of the stability region of hypothetical habitable Trojan planets," their investigations have shown that 7 out of 12 chosen exoplanetary systems can harbor habitable Trojan planets with stable orbits: HD 93083, HD 17051, HD 28185, HD 27442, HD 188015, HD 99109, and HD 221287. Here, they studied the stability regions using a numerical code based on a Bulirsch-Stoer integrator, and stability criterion being the maximum eccentricity of the Trojan as well as the Lyapunov Characteristic Indicator (LCI), integrating orbital periods of up to 10^7 periods of the primaries, and varying the normalized semi-major axis and the synodic longitude. Their work has resulted to a new catalogue of stable regions under different mass ratios between the primary (host star) and gas giant, as well as the eccentricities of their orbit. However, their work does not apply masses to the Trojan body, and only applied integration methods to lone gas giant planet systems.

In the work of Sudol and Haghighipour (2021) [17], they determined the transit-timing variations (TTVs) of hypothetical Trojan bodies in different Kepler circumbinary planetary systems: Kepler-16, -47, -453, -1647, and -1661. For integrations, they used the Bulirsch–Stoer Integrator in the MERCURY N-body integrator that has been developed specifically for integrating circumbinary planets. They found that stable Trojan planets are restricted to a narrow range of semi-major axes in all five systems and limited to small eccentricities in Kepler-16, -47, and -1661. TTV analysis indicates

that Trojan bodies fall within the detectable range of timing precision obtained from the Kepler telescope's long-cadence data, which points to Kepler data as a viable source to search for habitable Trojan planets.

Other studies in Trojan worlds do not always involve habitable conditions. In Kislyakova et al.'s work "On the ultraviolet anomalies of the WASP-12 and HD 189733 systems: Trojan satellites as a plasma source" [24], Trojan satellites are hypothesized to create anomalies in the spectrum of the gas giants in the WASP-12 and HD 189733 systems. Their study indicates that material outgasses from the molten surfaces of Trojan satellites in tadpole orbits around the Lagrange points L_4 and L_5. Their findings demonstrate that the orbital temperatures of WASP-12b are sufficient to melt the surfaces of rocky bodies, creating shallow lava oceans. For WASP-12b, this process results in the release of elements like Mg and Ca, which are anticipated to envelop the system, and for HD 189733b, the mechanism is only marginally possible due to the lower temperature. Their methods are done by investigating the long-term stability region of WASP-12b and HD 189733b in case of planar and inclined motion of these satellites and results show that unlike the classical exomoons orbiting the planet, Io-sized Trojans can be stable for the whole system's lifetime.

Multi-planet systems offer a wider range of variability in the stability of extra bodies in an exoplanetary system. In single-planet systems, the two bodies are the primary source of gravitational forces, which offers simplicity. However, multi-planetary dynamics can alter the course of other objects in the system [11].

In studying the stability of Trojans near gas giants, research has shown a relation-

ship between a gas giant's mass and the stable region for Trojan objects [15]. Generally, a higher mass gas giant can stabilize Trojans over a wider range of initial angular distances, due to the stronger gravitational pull around the Lagrange points. This stronger pull effectively enlarges the stability zone where Trojans maintain low eccentricities in the ”tadpole” orbits even at greater angular separations from the planet.

2.2 Gas Giants and Planetary Habitability

Gas giants have been one of the focus points in the search for planetary habitability. Although gas giants are inhospitable on their own, their large gravitational and thermal influences create environments where their natural satellites could become potential candidates for habitability. In the Solar System alone, two of the gas giants have already been speculated to have moons with certain characteristics of habitability. Several trips to the Jovian system have yielded intriguing knowledge about the exomoons [25] and has suggested that there are possible regions in Europa during its existence that could have been sites of biological activity. In the Saturnian moon Enceladus lies a similar case, where under a sheet of ice lies an ocean of water. Jets of water have been repeatedly detected to escape into space from the surface, as seen by the Cassini spacecraft [26]. Other moons such as Titan and Ganymede have also been popular locations for astro-biological research.

However, in the Solar System, these exomoons live around gas giants far from the Sun's habitable zone. Outside of habitable zone conditions, exomoons orbiting gas giants may experience tidal heating due to gravitational interactions, a crucial energy

source that can sustain subsurface oceans. For example, Jupiter's moon Io has geological activity due to tidal heating [27], and tidal heating is considered as a condition of planetary habitability [28]. But an intriguing population of exoplanets are gas giants that orbit around their host star's circumstellar habitable zone. This region has conditions that allow for liquid water on planetary surface, and a gas giant within this zone extends the possibility of habitable conditions to a collection of satellites. Studies in systems of Kepler planets have shown gas giants in habitable zones of their stars [29, 4, 17].

In studying planetary habitability, the habitable zone is a primary indicator of whether a planetary receives enough stellar radiation to allow for conditions for life [18]. Equation 2 shows the classical definition of the habitable zone is defined by the distance function that is dependent on the star's luminosity:

$$d = \left(\frac{L/L_{\odot}}{S_{eff}}\right)^{0.5} \text{AU} \tag{2}$$

where $L/L_{\odot}$ is the luminosity of the host star relative to the luminosity of the Sun, and S_{eff} is a constant called the effective solar flux. The luminosity L of a star is related to its radius R and effective temperature T_{eff} by the Stefan-Boltzmann law:

$$\frac{L}{L_{\odot}} = 4\pi\sigma \frac{R}{R_{\odot}}^{2} T_{eff}^{4} \tag{3}$$

where $R/R_{\odot}$ is the star's radius in solar radii, T is the star's effective temperature in Kelvin, and σ is the Stefan-Boltzmann constant, which has a value of $\sigma = 5.67 \times$

10^{-8} W $\cdot$ m^{-2} $\cdot$ K^{-4}. The primary determining factor in the habitable zone equation is the S_{eff}, where it represents the fraction of the stellar luminosity needed, compared to the solar luminosity at Earth's orbit. While other estimates of the habitable zone may incorporate different boundaries depending on stellar characteristics, this study focus on two well established values of the S_{eff} from the same study of Kumar Kopparapu et al. for the purposes of this estimation [30].

Table 2: Values used for the S_{eff} of determining the habitable zone boundaries.

Description	$\boldsymbol{S_{eff}}$	**habitable zone distance**
Recent Venus	1.7753	Inner habitable zone
Early Mars	0.3179	Outer habitable zone

However, it is important to note that the habitable zone is also not a rigid region and can vary depending on other parameters of the system [31]. While the potential for habitability exists, other astronomical factors pose challenges. For instance, intense radiation belts around gas giants could impact surface conditions. Studies and flybys of satellites to Jupiter alone has revealed intense radiation belts that could impact surface conditions and existence of habitability in Europa [32]. While moons remain the primary focus in discussions of habitability around gas giants, other bodies, such as Trojans—asteroids or small planets sharing a planet's orbit—also present unique possibilities. Many studies have started providing key insights in Trojans co-orbiting gas giants in the habitable zone [8]. Like moons, these Trojans may have stable orbits near the gas giant, and under certain conditions, they could also retain atmospheres or experience heating that might support habitability. Thus, in recent years, Trojans near gas

giants in the habitable zone have gained attention as potential hosts for life, broadening the scope of bodies worth exploring for signs of habitability. As observational methods improve, the study of gas giants and their moons will play an increasingly central role in astrobiology, offering new avenues for finding life-supporting environments beyond Earth.

2.3 Orbital Elements

Foundations of orbital mechanics were done by notable physicists Johann Kepler and Isaac Newton. In the context of orbital dynamics, any orbiting body will have several parameters measured. Kepler's laws of planetary motion (early 17th century) described how planets orbit the Sun, laying the groundwork for understanding orbital mechanics. However, the terminology used today became more formalized with the advancement of celestial mechanics in the 18th and 19th centuries. These parameters are also used in in simulating Trojan bodies to study their orbital dynamics [7, 8, 17]. In this study, the following orbital elements are of key importance [33]:

Semi-major Axis (a). The semi-major axis is the longest distance between the center and the edge of an ellipse. The concept of an ellipse with a semi-major axis comes from Kepler's first law, which describes the elliptical nature of orbits. In planetary motion, the semi-major axis is the longest distance between the star and the planetary body.

Eccentricity (e). Eccentricities describe the curvature of a conic section, with $e = 0$ describing a circle, $0 > e > 1$ describing an ellipse, $e = 1$ describing a parabola, and

$e > 1$ describing a hyperbola. Kepler introduced the idea of eccentricity to describe how ”stretched” an orbit is compared to a perfect circle. He used it in his laws to specify how ellipses differ from circles.

Inclination (inc). The inclination of an orbit refers to the tilt of the orbital plane relative to a reference plane. The term and concept of inclination were developed as astronomers described the three-dimensional orientation of orbits, especially as they mapped planetary motions.

True Anomaly (ν or f). The true anomaly is the angle between the direction of periapsis (the closest point in an orbit to the focus, such as a star or planet) and the current position of an orbiting object, measured at the focus of the orbit. This parameter was conceptualized to help describe the exact location of an object along its elliptical path, varying with time as the object moves. It is noted that the L_4 and L_5 Lagrange points are $f = 60°$ forward or behind an orbiting planet.

2.4 REBOUND and the WHFast Integrator

Planetary dynamics involve solving the equations of motion that determine the future positions of orbiting bodies. Since analytic solutions to N-body problems do not exist, numerical approximations are widely used to find stable solutions. One widely used method is the Wisdom-Holman method.

The Wisdom-Holman method [34] is based on the principles of symplectic integration, which preserves the Hamiltonian structure of dynamical systems. In symplectic integrators, energy-like quantities are conserved, and phase space volume is preserved,

leading to short-term accuracy when applied to orbital dynamics. The primary goal of the WHFast integrator is to find an approximate solution to the equations of motion for particles governed by the Hamiltonian $\mathcal{H}$ of the gravitational N-body system, written as the sum of kinetic and potential terms:

$$\mathcal{H} = \sum_{i=0}^{N-1} \frac{\mathbf{p}_i^2}{2m_i} \sum_{i=0}^{N-1} \sum_{j=i+1}^{N-1} \frac{G m_i m_j}{|\mathbf{r}_i - \mathbf{r}_j|} \tag{4}$$

where $\mathbf{p}$ is the momentum vector, and $\mathbf{r}$ is the position vector in Cartesian coordinates. The Wisdom-Holman method is specifically tailored for Hamiltonian dynamics and is designed for short-term integration of celestial mechanics problems, such as planetary orbits.

Rauch and Holman (1999), in ”Dynamical Chaos in the Wisdom-Holman Integrator: Origins and Solutions” [35], analyzed the nonlinear instability of the Wisdom-Holman integrator for highly eccentric orbits ($e > 0.9$), identifying artificial chaos caused by insufficient timestep resolution near periapse. They found that stability is achieved only with small timesteps. The study attributes this ”radial orbit instability” to overlapping step-size resonances and evaluates alternative methods. Among these, a time-regularized Wisdom-Holman mapping emerges as the most stable for eccentric orbits, while a refined potential-splitting method proves effective for systems with close encounters. Stark-based methods are deemed less practical for N-body integrations.

According to Viswanath (2002), in their study ”How Many Timesteps for a Cycle? Analysis of the Wisdom-Holman Algorithm” [36], in the Wisdom-Holman method, the timestep size is limited by orbital eccentricity; at least six steps per period are re-

quired even for circular orbits, with higher eccentricities demanding shorter steps. The method's application to solar system dynamics and other systems illustrates its strengths and constraints in capturing orbital behavior.

The paper "A study of symplectic integrators for planetary system problems: error analysis and comparisons" by Hernanzdez and Dehnen (2017) [37] analyzes and compares these approaches, highlighting the advantages of using Jacobi coordinates and the Hernandez (H16) method, a 16th-order symplectic integrator primarily used in numerical integration for Hamiltonian systems. These methods avoid issues like center-of-mass inaccuracies and truncation errors found in other approaches. Using H16, the study computes Pluto's orbital evolution over 500 million years in a new calculation, showcasing its precision and reliability.

Rein and Tamayo created the WHFast integrator based on the WH symplectic integrator [38]. WHFast is ideal for systems with a dominant central body and minor perturbations to Keplerian orbits. The method involves constructing a symplectic integrator by splitting the Hamiltonian into smaller, more easily integrated parts. WHFast employs an adaptive time-stepping algorithm, which is crucial for managing computational resources while maintaining accuracy. The timestep can be increased when the system is stable and decreased during rapid dynamical changes. To handle close encounters between bodies, WHFast uses specific criteria to switch to a smaller timestep, such as checking the distances and velocities of bodies, ensuring accurate integration during critical interactions. Further details on the integrator's programming are discussed in Rein and Tamayo's paper "WHFast: A fast and unbiased implementation of a

symplectic Wisdom-Holman integrator for long term gravitational simulations" [38].

Like all integrators, errors are present due to various factors in their application. The implementation of WHFast introduces three primary sources of error. First, there is an inherent error in the integrator itself, as it does not perfectly solve the equations of motion for the Hamiltonian $\mathcal{H}$. Second, errors arise from the limitations of finite numerical precision in computer representations. Third, if any components of the integration algorithm introduce bias, the resulting errors become correlated, leading to a faster long-term growth of energy error compared to uncorrelated errors. This growth scales as $E_{bias} \sim N \sim t$. Which of these errors dominates depends on factors like the integrator used, the timestep size, and the total integration duration (i.e., the number of timesteps).

The REBOUND software package, primarily used for this study's simulations [39], integrates the motion of particles under gravitational influences. It can also apply integration methods to arbitrary user-defined ODEs coupled to N-body dynamics, such as tides, spin, or other forces. The implementation of thepackage is described in Rein and Liu's paper "REBOUND: an open-source multi-purpose N-body code for collisional dynamics". REBOUND was developed for studying collisional dynamics, like those in planetary rings, but is also capable of solving the traditional N-body problem. Its highly modular structure allows for easy customization, making it adaptable to a broad range of problems in astrophysics and other fields.

REBOUND has been used in over 600 research publications in various astronomical studies. The Integrator with Adaptive Step-size control, 15th order (IAS15) integrator

within REBOUND was used to simulate circumbinary habitable zones analyzed with machine learning techniques in Pytorch by Kong et al. [40]. Their research focuses on circumbinary (P-type) planets and analyzes habitable zones based on planetary trajectories and dynamics. Key factors influencing orbital stability and habitability include the binary stars' mass ratio and orbital eccentricity. The study categorizes planetary habitability into habitable, periodically habitable, and non-habitable zones and uses a machine-learning model to efficiently classify these systems.

Dong et al. [41] focus on TOI-2145b, a 5.68 M_J planet with a 10.26-day orbit ($e = 0.214$) around an evolved A-star. Data from TESS, Keck/HIRES, and the NEID spectrometer were used, with the projected stellar obliquity constrained to $\lambda = 6.8$. N-body simulations using REBOUND and the IAS15 integrator suggest either high initial mass or collisional growth as plausible formation scenarios, with the latter explaining the observed mass-eccentricity trends in warm Jupiters.

2.5 Chaos Indicator: MEGNO

In the context of astrodynamics, stability criteria are prominently used alongside integration methods. This study utilizes the Mean Exponential Growth of Nearby Orbits, or MEGNO, a dynamical indicator used to assess the chaotic nature of an orbit in a dynamical system, such as planetary orbits. This indicator is relatively new, introduced formally by Cincotta et al. in their paper "Simple tools to study global dynamics in non-axisymmetric galactic potentials - I" [42]. It provides a measure of how nearby orbits diverge over time, helping to distinguish between regular (stable) and chaotic

(unstable) behavior. MEGNO is a scalar function of time. In an N-body simulation, MEGNO calculations start by tracking the evolution of a reference orbit and a nearby ”perturbed” orbit. This perturbed orbit is slightly offset from the reference, creating a pair of ”nearby” trajectories. The MEGNO algorithm computes a ”deviation vector” between these two orbits, often denoted as $\delta(t)$. This vector shows how much the perturbed orbit has drifted from the reference. Over time, the length of the deviation vector grows, often exponentially in chaotic systems. MEGNO captures this growth by evaluating $\ln \delta(t)$ over time. The goal is to measure how quickly (or slowly) nearby orbits diverge. MEGNO uses a specific averaging factor, $\bar{\gamma}$, to reflect the nature of this divergence. It evaluates how the distance grows in terms of a time-weighted average

MEGNO is derived from the Lyapunov Characteristic Number (LCN), a stability indicator widely used in astrodynamics. The LCN is a powerful tool for analyzing the long-term behavior of Hamiltonian systems. In chaotic systems, trajectories diverge exponentially over time, resulting in a finite Lyapunov timescale. Consequently, calculating the LCN helps determine whether a system is chaotic and, if so, provides an estimate of the associated timescale [38]. While the LCN is effective for identifying chaos, it does not provide detailed insights into the structure of regular (non-chaotic) components. The LCN, σ, of a given orbit, $\gamma(t)$ is defined mathematiaclly as (from Eq. (21) of [42]) :

$$\sigma(\gamma) = \lim_{t\to\infty} \frac{1}{t} \ln \frac{\delta(\gamma(t))}{\delta_0} \tag{5}$$

where δ_0 are infinitesimal displacements from γ at times t and 0. Formally, the MEGNO

, denoted as $\mathcal{J}$, of a system is described by a function based on the LCN (from Eq. (22) of the same paper):

$$\mathcal{J}(\gamma(T)) \equiv \frac{2}{T}\int_0^T = \frac{\dot{\delta}(\gamma(t))}{\delta(\gamma(t))}\, t\, dt \tag{6}$$

where $\gamma_r(T)$ is the given orbit, $\delta(\gamma_r(T))$ are infinitesimal displacements from γ at time t. The following table describes orbital dynamics for the values of MEGNO [43]:

Table 3: MEGNO values and their interpretations.

MEGNO Value	**System Description**
$Y(t) \to 2$	regular (quasi-periodic or stable)
$Y(t) \to \infty$	chaotic (exponential divergence)

Should the LCN be needed, it can be recovered from the MEGNO via a linear least square fit to $\mathcal{J}(T)$ for any T.

The REBOUND environment has an in-built access to different stability criteria such as Lyapunov numbers and MEGNO. Although there isn't an exact boundary of values for the MEGNO, it is described that as long as $\mathcal{J}$ is constant around 2, regular motion is taking place, while irregular motion outputs MEGNO value that grows linearly with time with a rate that is the LCN. Further description of the MEGNO is in the paper of Cincotta et al. [42].

MEGNO has been used and verified multiple times by other studies. Goździewski et al. (2001) applied the indicator to the planetary system υ Andromedae system and

have claimed that their work demonstrates the usefulness of the MEGNO in planetary dynamics [44].

Maffione et al. tested MEGNO in a study of a sample of orbits in a triaxial galactic system, previously studied using another stability criteria, the Finite Time Lyapunov Characteristic Numbers (FT–LCNs) by other authors, and have shown a good correlation between MEGNO and FT-LCN values for different integration times, indicating similar strength in determining regular versus chaotic motion [45]. To note, the FT-LCN is a localized version of the LCN that evaluates the divergence rate over a finite time interval. The FT-LCN provides insight into the system's behavior over shorter time scales and is particularly useful for identifying transient dynamics, detecting regions of phase space with varying stability, and studying systems where infinite-time integration is impractical.

Hinse et al. (2010) applied the MEGNO technique to the dynamics of irregular satellites of Jupiter, and have hypothesized on the possibility of gravitational scattering from high-order mean motion resonance as a dynamical cause to explain the observed orbital velocity dispersion for members of the Pasiphae family [43].

Cincotta et al's paper "Phase space structure of multi-dimensional systems by means of the mean exponential growth factor of nearby orbits" [46] explores the use of the MEGNO as an alternative method for studying global dynamics in Hamiltonian systems. They have stated that MEGNO is effective in investigating both regular and chaotic components of phase space, revealing resonance structures, stable and unstable periodic orbits, and measures of hyperbolicity in chaotic regions, similar to the

Lyapunov characteristic number. The technique is applied to a simple model, the 3D perturbed quartic oscillator, to visualize its phase space and resonance structure, with additional examples demonstrating its use in multi-dimensional canonical maps.

2.6 NASA Exoplanetary Archive

The NASA Exoplanet Archive is an extensive online database that provides a wealth of observational data on confirmed exoplanets and their host stars, curated to support exoplanetary research. It includes a variety of datasets, such as stellar parameters, orbital elements, and detection methods, and offers visualization and analysis tools. Data is cllected from all kinds of astronomical studies on space missions that use satellites that use a multitude of instruments to collect data on different astronomical studies. Researchers can access detailed information on exoplanet masses, radii, and orbital characteristics, as well as host star properties like spectral type, luminosity, metallicity and effective temperature, which are crucial for astrophysical modeling, such as planetary environments, gravitational stability, and stellar dynamics. The archive is an essential resource for validating simulation models and informing studies of exoplanetary stability and dynamics.

Chapter 3

METHODOLOGY

3 Methodology

3.1 Experimental Design

To analyze short-term Trojan stability in gas giant orbits that are in the habitable zone of multi-planet systems, this study will use the REBOUND software, equipped with the WHFast numerical integrator, and MEGNO calculator, using data from the NASA Exoplanetary Archive. This study will focus on exoplanetary systems that have gas giants in the habitable zones of multi-planetary systems. All results were saved in Comma-Separated Values (CSV) files. Data preprocessing and filtering, experimental simulations, and post-simulation data collection and analysis were performed in the Scientific Python Development Environment (Spyder) environment. Figure 5 shows the flow of the experimental process in determining initial parameters of stable Earth-mass Trojans in gas giants in the habitable zones of multi-planet systems, and Figure 6 describes the flow of the algorithm used in simulating and determining the MEGNO values.

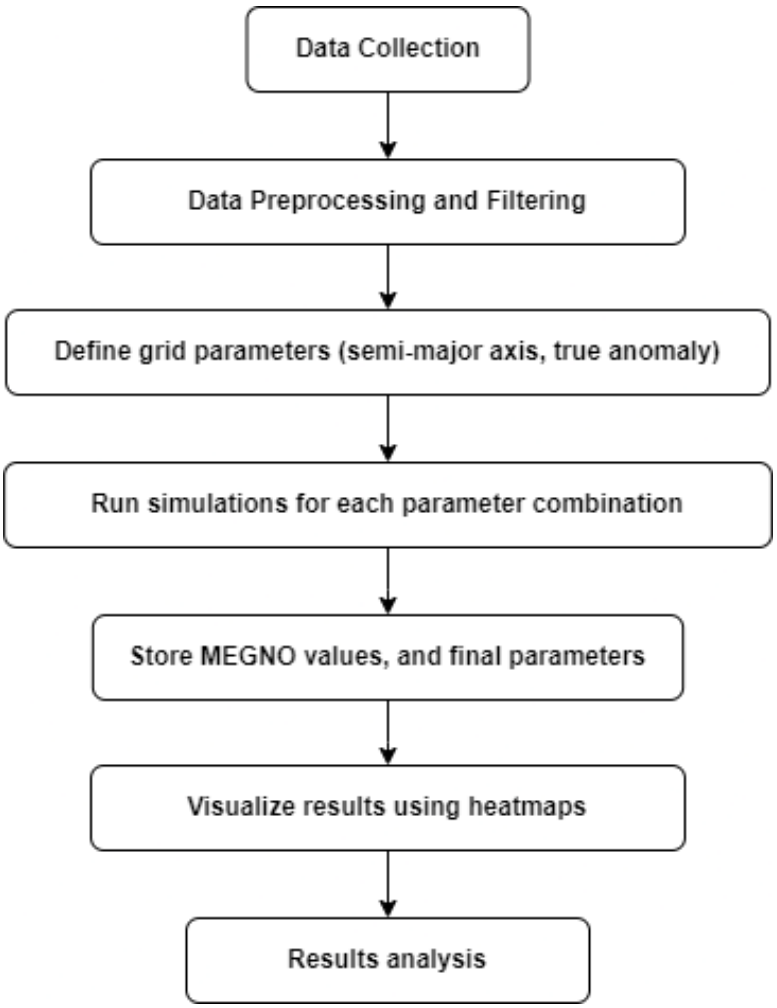

Figure 5: General overview of the experimental design and methodology.

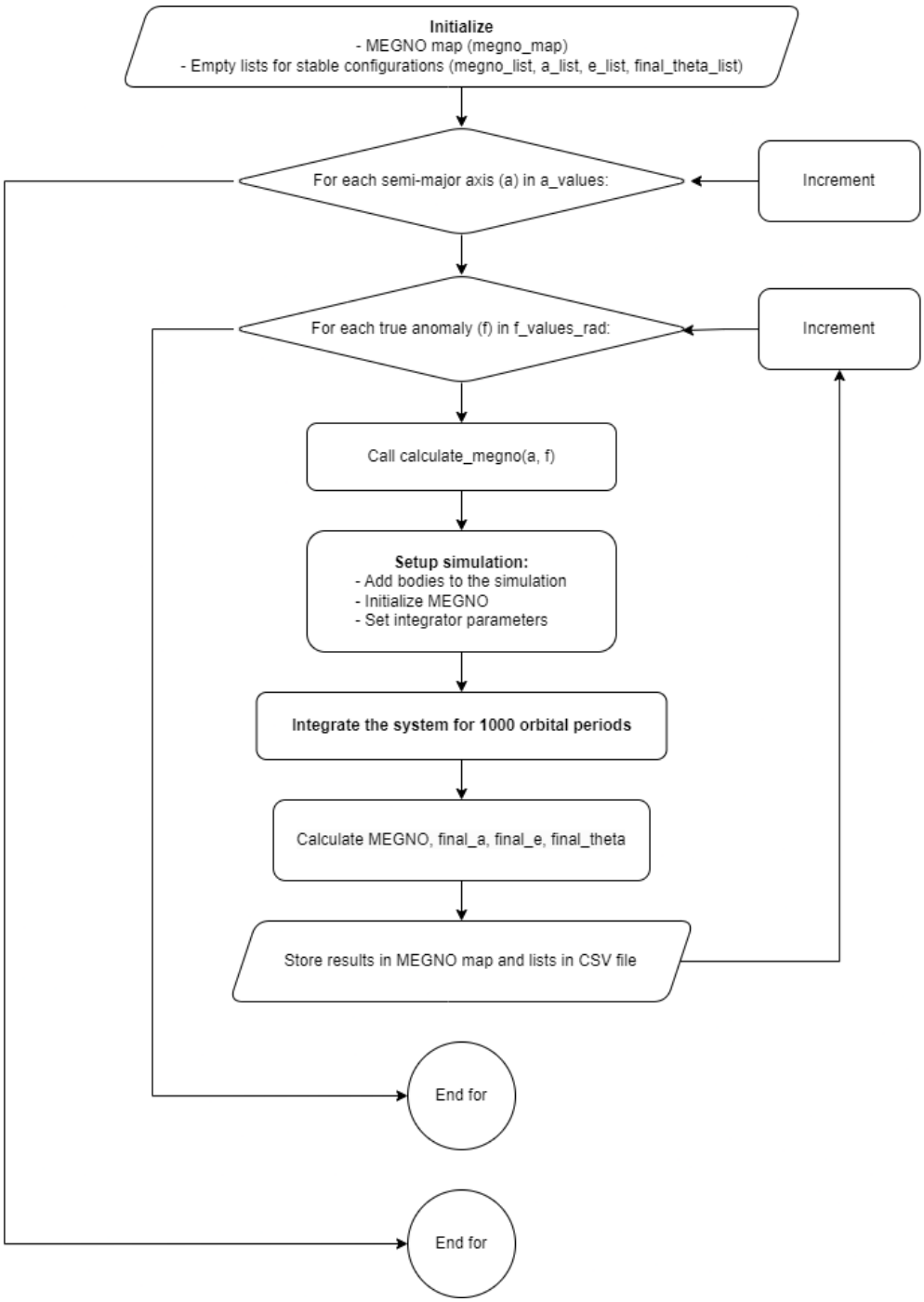

Figure 6: Flowchart of the algorithm used in simulating and storing outputs.

3.2 Data Selection and Filtering

The exoplanet data used in this study comes from the NASA Exoplanet Archive, which provides parameters for both confirmed exoplanets and candidates, along with the associated references. To ensure suitable candidates for the simulations, the complete list of exoplanetary system will be pre-processed based on specific criteria, detailed below.

Single Host Star, Multi-Planetary Systems. Exoplanetary systems with a single host star and multiple orbiting planets will be selected for this study. This approach focuses on systems with more than one planet to investigate the gravitational interactions between multiple bodies. The choice is motivated by the need to simulate realistic dynamical environments where stability analyses are meaningful. Multi-planetary systems offer diverse configurations that can provide valuable insights into stability mechanisms.

Small Eccentricities. A range of $e < 0.25$ will be chosen as the defining limit of orbit stability, as higher eccentricities often lead to unstable orbital configurations and significant gravitational perturbations between planets. High-eccentricity orbits are prone to planet-planet scattering, which can disrupt stable configurations [47]. Additionally, [8] have noted that eccentricities greater than 0.3 allows the planet and therefore the Trojan to orbit outside the habitable zone, especially for those orbiting near the boundaries of the habitable zone. Thus, filtering for low eccentricities ensures a focus on more dynamically stable systems that are also more likely to stay inside the habitable zone.

Gas Giant Mass Range. Mass ranges of $0.3\ M_J < m < 13\ M_J$ will be filtered.

This range accounts for gas giants that are substantial enough to influence the dynamics of hypothetical Trojan bodies but are not massive enough to be considered brown dwarfs. This selection aims to simulate systems where gas giants play a dominant role in the orbital stability of surrounding objects. The exact ranges are based on existing studies on hypothetical Trojans in exoplanetary systems.

Inside the Host Star's habitable zone The planet's host stars were checked for luminosities, and missing values were solved using the Stefan-Boltzmann law using stellar radius and effective temperature values. This method follows previous approaches, such as those outlined by [17], who analyzed the habitable zones in binary star systems. Here, the habitable zone boundaries will be computed using established equations, and gas giants with semi-major axes that fall within these boundaries will be retained for further analysis.

Handling Missing Data and Uncertainty. For stars lacking radius or temperature information, the Stefan-Boltzmann law will be applied only when reliable estimates are available. If essential parameters are missing and cannot be reasonably inferred, those systems will be excluded. The method assumes precise and consistent stellar parameter values, acknowledging potential limitations in the data.

The data pre-processing steps will be implemented in Python, primarily using the Pandas and NumPy libraries. Here's a breakdown of the procedure:

1. **Initial Filtering:** The code first filters the data using the default parameters to ensure that only the primary parameters (indicated by 'default_flag' == 1) are used. This is because the archive collates data from different studies of the same

system. This filtered data is saved in a CSV file.

2. **Sequential Filtering for Simulation:** Next, additional filters are applied sequentially to narrow down the dataset further:

   (a) Single host stars: (`sy_snum == 1`)

   (b) Systems with more than one planet: (`sy_pnum > 1`)

   (c) Planetary mass between 0.3 and 13 M_J: (`0.3 < pl_bmassj < 13`)

   (d) Eccentricity less than or equal to 0.25: (`pl_ecc` $\leq$ `0.25`)

   The resulting filtered data is saved in another CSV file.

3. **Luminosity Calculation and Habitable Zone Filtering:** The luminosity of each star is calculated using the Stefan-Boltzmann law. The calculated luminosity values will then be used to determine the inner and outer boundaries of the habitable zone, applying the equations from well-established models. Planets will be checked for whether their semi-major axes lie within these boundaries, and this information will be added as a new column. This subset is saved into another CSV file.

4. **Manual Check for Companion Planets:** Since the filtering process might exclude some planets from multi-planetary systems, a manual check will be performed using the original dataset to ensure all companion planets in selected systems are accounted for. This step is necessary to provide a comprehensive dataset for analysis. Potential biases from this manual check will be noted as a limitation,

as it introduces some subjectivity into the data selection process. This final list will be saved into a final CSV file.

Manual checks for multi-planet systems are performed to ensure completeness of the dataset. To reduce subjectivity, we cross-referenced the selected systems with independent datasets, where available, and documented the criteria for rejecting ambiguous systems. Despite these efforts, this manual step remains a limitation that could be addressed with automated algorithms in future studies.

3.3 Simulating Trojans

3.3.1 Simulation Framework

Simulations in this paper made use of the REBOUND N-body code [39]. The simulations were integrated using WHFast, a symplectic Wisdom-Holman integrator [38, 34]. Variational equations were used to calculate trajectories of nearby orbits [48].

3.3.2 Model Setup

The parameters for the simulations are sourced from the NASA Exoplanet Archive. Key details for both host stars and exoplanets are considered:

Stellar Parameters: Mass (used directly in the N-body simulations), radius, and temperature (used to compute luminosity and define the habitable zone boundaries).

Planetary and Orbital Parameters: These include mass, semi-major axis (relative to the host star), eccentricity, and inclination.

Trojan Parameters: Mass will be that of Earth's ($M_{Earth} = 3 \times 10^{-6}\ M_{\odot}$), initial

semi-major axis, eccentricity, and inclinations will be equal to that of the selected host gas giant in the habitable zone.

Initial conditions for the simulations are set based on these parameters, with REBOUND handling the positioning and dynamics setup of the planetary bodies.

3.3.3 Simulation Procedure

The simulations will use WHFast, a symplectic integrator ideal for short-term integrations of systems dominated by a central body. While WHFast efficiently handles Keplerian dynamics, it may encounter challenges in highly chaotic systems or during close encounters. To ensure accuracy, the integration timestep is adjusted dynamically, and results are validated by cross-checking key outcomes against prior studies, such as Schwarz et al. [8]. The simulations are designed to run for a duration equivalent to 1,000 orbits of the host gas giant, or the gas giant that will have the Trojan co-orbiting body, corresponding to a short-term investigation of orbital stability. This timeframe is chosen to allow for the detection of potential instability mechanisms, such as close encounters or ejections, and to establish whether the system exhibits short-term stability before longer-term integration becomes computationally expensive. However, it is important to note that a short-term analysis of orbits does not fully determine stability, work by Schwarz et al. has determined that integration time of 0.1% (10^4 periods) provides a good first approximation, but the stability region may be overestimated by up to 20%. The timestep of the simulation is 1% of the orbital period, which is well inside the minimum timestep for minimal energy errors. It is also the recommended timestep

by other related studies [8, 17].

The simulation will investigate a range of initial configurations, including:

1. **Earth-Mass Trojans:** To analyze the stability of hypothetical Earth-mass Trojan bodies in the system.

2. **Eccentricity and Inclination of Host Planet:** The initial eccentricity and inclination of the simulated Trojan is equal to that of the co-orbiting host planet to ensure that any changes in the orbital stability of the Trojans were due to variations in the semi-major axis and angular distance, while the conservation of angular momentum was maintained throughout the runs.

3. **Varying Semi-Major Axes:** Trojans will be placed within 5% of the gas giant's initial-semi-major axis distance. This is so that the gravitational forces from the gas giant are large enough to be considered as Trojans, and not as new planets.

4. **True Anomaly Range:** Initial true anomaly values will vary from 30° to 90° to evaluate the gravitational effect of distance from the gas giant.

5. **Gravitational Forces Only:** The simulations will only consider gravitational interactions, excluding non-gravitational forces such as radiation pressure or tidal effects, to isolate the impact of gravity.

6. **Different Number of Planets.** The simulation will be repeated n times for each system corresponding to the number of planets that are found in the system. Each simulation run will have n number of planets, starting from the host gas giant, and is added in order of the distances of the planets.

3.3.4 Data Collection and Analysis

The primary data to be collected from each simulation run will include:

Orbital Elements. Key orbital parameters such as eccentricity and angular distances will be recorded after each simulation. A heat map of the initial parameter values (eccentricity and angular distance) vs the output parameters (orbital elements) will be generated for analysis.

Stability Indicators. This study will use the stability criteria MEGNO to quantify the chaotic nature of the orbits. A heat map of MEGNO values will be generated for different semi-major axes and true anomalies of the Trojans.

As an additional note, the algorithm will automatically check for instability by setting a maximum exit distance. If a simulation detects this, the resulting MEGNO and final parameter values will be set to 0. The results will be analyzed by comparing the stability performance across different systems and initial conditions. The analysis will focus on identifying trends and patterns that govern short-term stability or instability in Trojan configurations in the context of multi-planetary systems. To ensure numerical accuracy, consistency with prior work, and reliability of findings, validating these results are necessary. Validation steps include numerical analysis (determining the fraction of the portion and centroid of the stable region), comparing stability trends with known cases from prior studies (such as by Schwarz et al. (2007) [8] and Sudol et al. (2021) [17]), and checking for consistency with theoretical expectations. A detailed comparison of the findings with established works is presented in the Results and Discussion section.

Chapter 4

RESULTS AND DISCUSSION

4 Results and Discussion

From the data processing, twelve exoplanetary system have satisfied the criteria for simulations. However, due to the extent of the computational power available, only two systems will be explored in detail. It is recommended that future research apply the methods used here in the unused systems. The selected exoplanets were based on the number of planets as well as the variation in the parameters (masses, semi-major axis, eccentricity, inclination). Table 4 below details stellar parameters of each system as well as habitable zone distances.

Table 4: Stellar parameters and habitable zone distances of selected exoplanetary system.

System	**Mass** $[M_\odot]$	**Radius** $[R_\odot]$	**Effective Temperature [K]**	**Luminosity** $[L_\odot]$	**Inner habitable zone [AU]**	**Outer habitable zone [AU]**	**Ref.**
47 UMa	1.06	1.21	5872	1.568	0.94	2.22	[49]
HD 108874	1.10	1.05	5600	0.977	0.74	1.75	[50]

For each Trojan planet, an integration of the five-body (for 47 UMa) or four-body (for HD 108874) system of star-planets-Trojan for 1000 orbital periods of the host gas giant situated in the habitable zone. The timestep of the integration was 1% of the orbital period. The algorithm stops an integration after the total time period, or once a particle in the system is ejected, leaving behind a high MEGNO value. To check if the simulation software is working properly, an initialization test is done in the 47 UMa system, where one Trojan planet was simulated, and its orbital path is traced out.

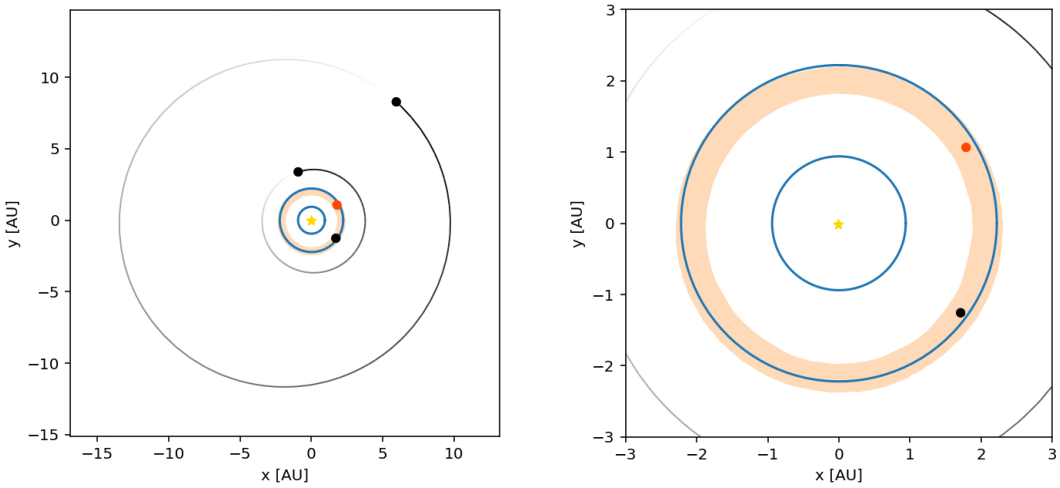

Figure 7: Overview of the 47 UMa system after 10^3 periods.

Figure 7 shows the orbital trail of a simulated Trojan after 10^3 orbital periods with initial parameters $a = 2.1$ AU, $e = 0.032$, $inc = 0°$, with the entire system (left panel) and a close-up of the inner host gas giant and Trojan orbits (right panel). The yellow asterisk is the primary, the three black dots are the planets 47 UMa b, 47 UMa c, and 47 UMa d, respective to the distances. The orange region is the orbital trail of the Trojan, and the blue concentric circles are boundaries of the habitable zone. In this run, the Trojan remains inside the habitable zone for an extended period. Although it seems that the Trojan body escapes this area, as stated previously, the habitable zone is not a fixed boundary. Next, a visualization of the distribution of Trojans to be simulated is shown.

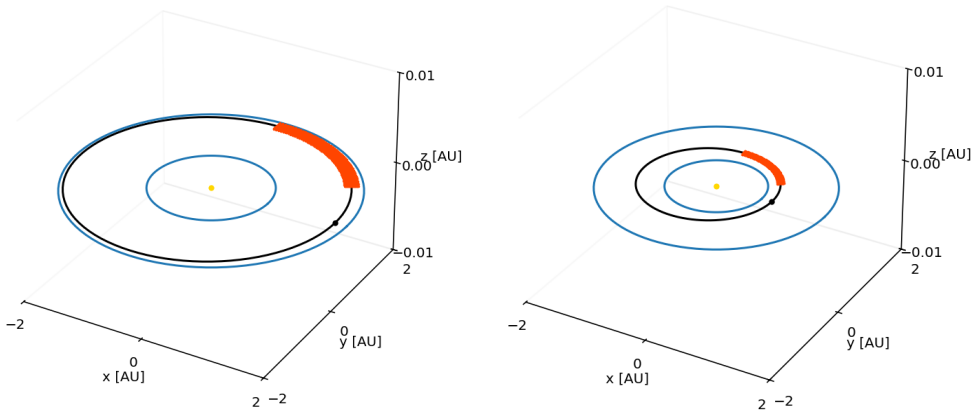

Figure 8: Spatial distribution of hypothetical Trojan planets in the orbit of selected planetary systems' host planets.

Figure 8 shows the distribution of each simulated Trojan in the 47 UMa b and HD 108874 b orbits with given initial semi-major axis and true anomaly. The blue concentric circles are the boundaries of the habitable zone. The positions of the simulated Trojans are along the orbit of the host gas giant, varying in angular distances ($30° > f > 90°$) from the gas giant.

The following sections will discuss the results for each planetary system 47 UMa and HD 108874.

1. 47 Ursa Majoris System

Table 5: Initial orbital parameters of planets in the 47 UMa system.

Planet	**Mass** M_J	a **[AU]**	e	inc **[deg]**	**Ref.**
47 UMa b*	2.53	2.1	0.032	0	[49]
47 UMa c	0.54	3.6	0.098	0	[49]
47 UMa d	1.64	11.6	0.16	0	[49]

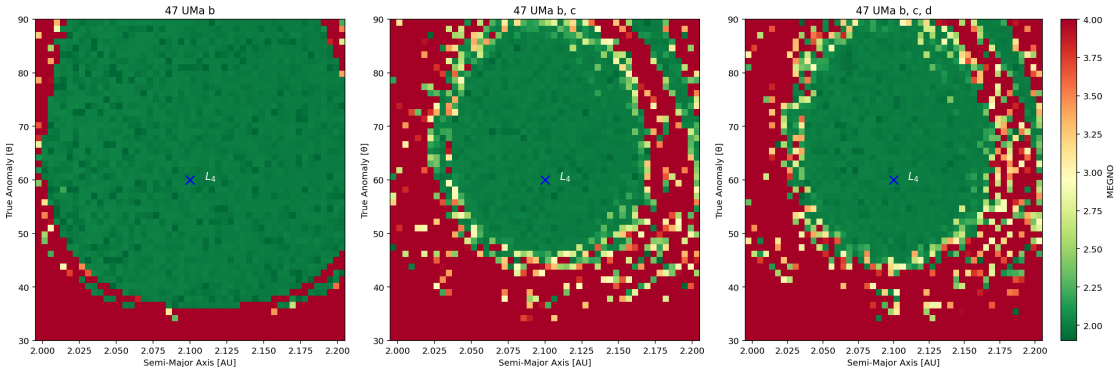

Figure 9: MEGNO stability maps for varying initial semi-major axis (a) and true anomaly (f) values across different planetary configurations in the 47 UMa system.

In the 47 UMa system, plotting the MEGNO heat maps for different n planets (Figure 9), all have shown stability regions in the parameter space of the initial semi-major axis and true anomalies. In these heatmaps, shown by Figure 8, MEGNO values close to 2 are green, while MEGNO values greater than two range from yellow to red, and the location of the Lagrange point is shown as a blue x-symbol. The green region is therefore the stable region, and yellow and red regions correspond to regions of instability.

This is also present in previous works of Schwarz et al., where the LCN map of HD 221287 was shown to have a stability region around the L_4 point. For all systems, the stable regions (where MEGNO ≈ 2) are approximately centered in the Lagrange point. Similarly, in Schwarz's work [12], which only consisted of single planets in the planetary system, there is a small area outside of the region where it shows that those parameters provide stable orbits, however this is an approximation error, and is resolved in higher timescale integration. In the boundaries of the stable region, there are distortions in the data, suggesting instability or being highly sensitive to changes in the parameters. Nonetheless, the results prove that the area around the Lagrange point

can host a Trojan planet, even in multi-planetary system conditions.

One apparent result is that the stability region shrinks from $n = 1$ planet to $n = 2$ planets. Considering that the integration only consists of gravitational forces, the shrinking can be attributed to the effects of the close neighboring planet, 47 UMa c. The distance from 47 UMa b to 47 UMa c is $a = 3.6 - 2.1 = 1.5$ AU. Moreover, the third planet, 47 UMa d, seems to have little effect on the stability region, which can be attributed to the distance ($a_{47\ UMa\ d} = 11.6$ AU).

Comparing this result to Trojan asteroids in the Solar System, we see that the possible orbital range of the Trojans is relatively the same. In Jupiter, the asteroids are around 3-4% varying in distance around Jupiter's distance from the Sun [43], which is close to the range of distances that allow for stable orbits in the Trojans around the selected multi-planetary systems.

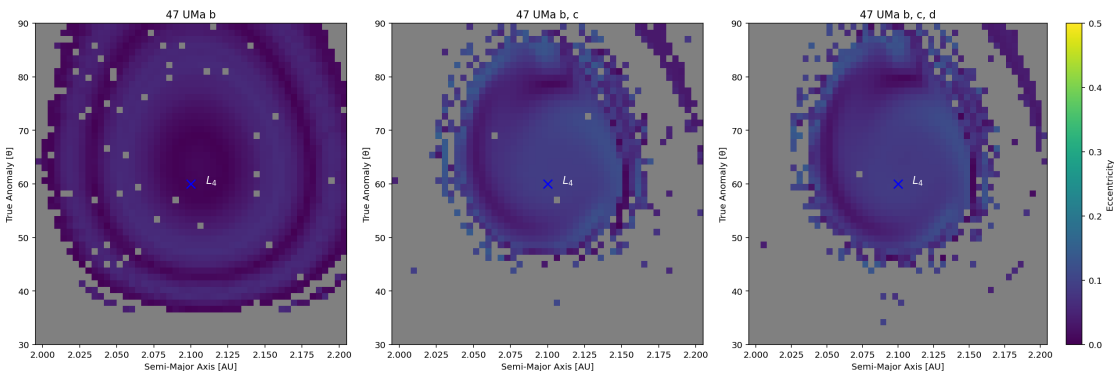

Figure 10: Final eccentricity maps for stable configurations of initial a and f values across different planetary configurations in the 47 UMa system.

As done by previous studies [13], this research will use the eccentricity as another criteria of stability. The heat maps of the final eccentricities of each stable simulated Trojan planet are shown in Figure 10. Dark blue colors represent low eccentricities,

$e \approx 0$, and green to yellow values represent $e \geq 1$. The gray region corresponds to the unstable region from the MEGNO plot.

The size of the stable region is consistent with the corresponding MEGNO plot. The central stable region is centered around the Lagrange point, as is the case of the MEGNO. In the stable region, simulated Trojan planets end in low eccentricities after the integration, suggesting low deviation of orbits. This is also shown to previous works by Schwarz [8] and Sudol [17] in using maximum eccentricity as stability criteria for the orbits, and further discussion is provided after the findings.

It is also evident in the plots that there are symmetries in the final eccentricities of the parameter space. These concentric rings correspond to points of symmetries, where simulating the initial parameters corresponding to these rings will output similar values, which implies a symmetry in orbital dynamics around the Lagrange point, due to the way orbital positions repeat after a certain period or distance.

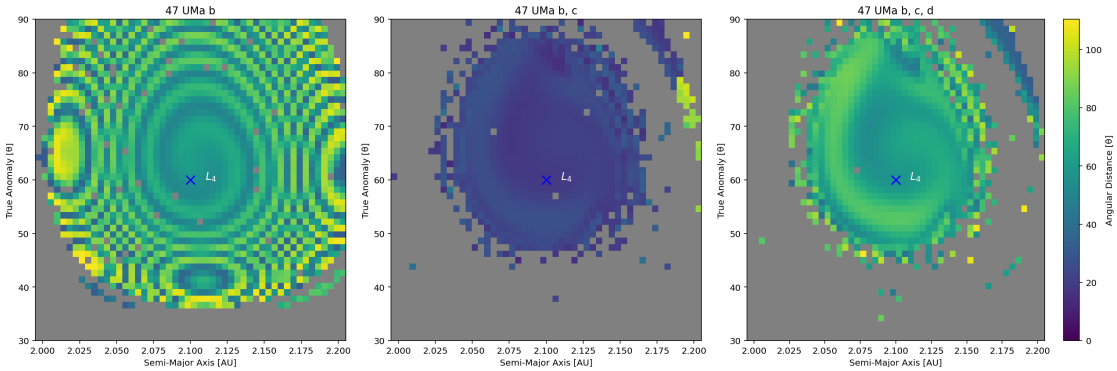

Figure 11: Final angular distance maps for stable configurations of initial a and f values across different planetary configurations in the 47 UMa system.

The final parameter to be analyzed is the distances between the Trojan body and the host gas giant. Figure 11 shows the plots for the final distances of the Trojan planets in

terms of the angular distance between the Trojan and the host gas giant. For reference, blue values are for $\theta < 50°$, green values represent $50° \geq \theta \geq 80°$ and yellow for $\theta \geq 90°$, and gray areas are regions of instability based from the MEGNO map.

Similarly, the size of the stable regions is consistent with the corresponding MEGNO plots. The central stability region is also present in the parameter space, as well as the symmetry of the output values. In the angular distance, the final values do not seem to be consistent in every initial input. Every initial input of angular anomaly can produce different final angular distances. Rings of similar values may suggest that the Trojan's angular separation oscillates around an equilibrium value outside of the Lagrange point. The variation in true anomaly and semi-major axis affects the amplitude and frequency of these oscillations, creating zones with consistent final angular distances. Notably, for the simulations under a single planet, the final angular distance can vary wildly and has several points of symmetry. However, these points become less visible, and the stable region produces a more consistent final angular distance for $n = 2, 3$ planets in the system. For $n = 2$ planets, final angular distance ranges in lower values, around $\theta < 50°$, suggesting the Trojan is librating in the region around the Lagrange point. For $n = 3$ planets, even if the third planet is relatively far away, the Trojan planet ends up closer to the Lagrange point, at around $\theta \approx 60°$.

Therefore, for the 47 UMa system, analysis of short-term dynamics conclude that hypothetical Trojan planets can co-exist with the gas giants around the Lagrange point L_4, with more stable orbits if considering both planets 47 UMa b and 47 Uma c.

2. HD 108874 System

Table 6: Initial orbital parameters of planets in the HD 108874 system.

Planet	**Mass** M_J	a **[AU]**	e	inc **[deg]**	**Ref.**
HD 108874 b*	1.42	1.04	0.13	0	[50]
HD 108874 c	0.99	2.659	0.239	0	[51]

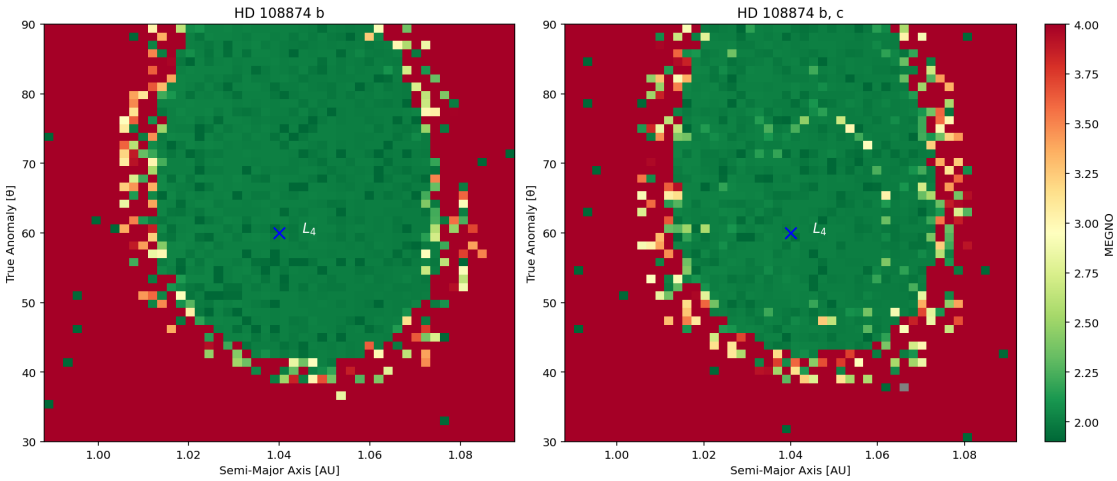

Figure 12: MEGNO stability maps for varying initial semi-major axis (a) and true anomaly (f) values across different planetary configurations in the HD 108874 system.

In the HD 108874 stability region heat map, shown in Figure 12 , the size of the stability region is identical when either $n = 1$ or $n = 2$ planets are simulated. Likewise, the stability region is centered around the L_4 Lagrange point of HD 108874 b. Therefore, it can be concluded that similar systems could host a hypothetical Trojan planet.

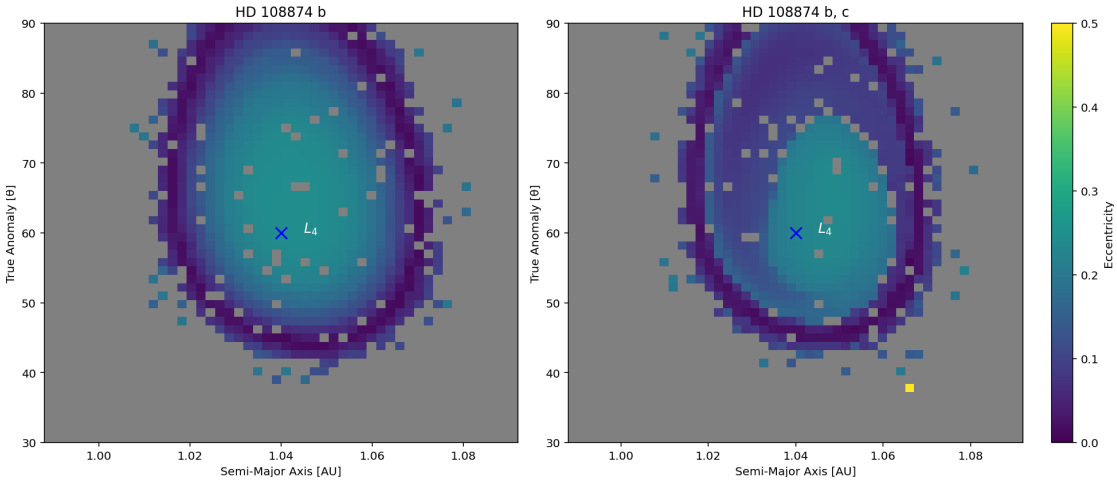

Figure 13: Final eccentricity maps for stable configurations of initial a and f values across different planetary configurations in the HD 108874 system.

For the final eccentricity maps for the stable configurations, the simulated Trojans end up in low eccentricities, mostly around $e \approx 0.2$. This is higher than the host gas giant's eccentricity. Therefore, the eccentricities of the simulated Trojans have changed throughout the integration. It can also be noticed that for $n = 2$ planets, or for the entire system, the simulated Trojans seems to have lower final eccentricities, suggesting stability in that orbital parameter.

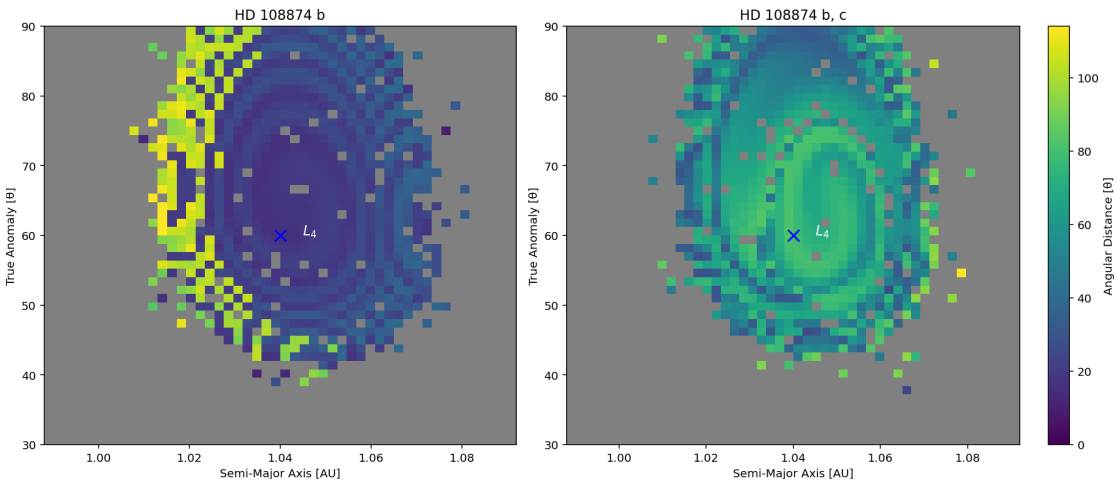

Figure 14: Final angular distance maps for stable configurations of initial a and f values across different planetary configurations in the HD 108874 system.

The final angular distance maps for the stable regions show a deeper story. For $n = 1$ planet, there are initial parameters that pulls the Trojan away from the Lagrange point, shown in the figure as the yellow region corresponding to angular distances $\theta > 80°$. This implies that, despite orbital stability, the simulated Trojans are in erratic motion, moving away from the Lagrange point. It can be deduced that further integration could show that the orbits can or cannot be unstable. For $n = 2$ planets, the final angular distances are relatively close around $\theta \approx 60°$, which meant that the Trojans librate around the Lagrange point at more stable values.

It can be then further concluded that for the HD 108874 system, the 2-planet model is much more stable than a single planet model and can host more stable hypothetical Trojan planets.

The following section will discuss the numerical analysis of this study. The fraction of the stable region (indicated by MEGNO) compared to the parameter space as well as the centroid of the stable region are measured for all configurations.

Table 7: Fraction and centroid of the stable region in the parameter space heat maps of the MEGNO stability criteria.

Planet		Stable Fraction	Centroid Point (a, f)
47 UMa	1 planet	0.774	(2.106, 64.096)
	2 planets	0.4024	(2.103, 67.624)
	3 planets	0.414	(2.103, 67.573)
HD 108874	1 planet	0.3812	(1.043, 66.689)
	2 planets	0.2636	(1.041, 68.886)

In the 47 UMa system, the fraction of the stability region decreases from $n = 1$ planet to $n = 2, 3$ planets, from 77% to $40 - 41\%$. For $n = 3$ planets, the size of the region seems to be as much as high or even slightly higher than when $n = 2$ planets

are involved. This could be attributed to the external gravitational influence stabilizing the region, or to the numerical setup of the simulation, which may lead to inaccurate approximations of the orbits. In reality, the potential profile of the N-body Hamiltonian is such that the Lagrange points L_4 (and L_5) are saddle points, where the gravitational forces from the two primary bodies create a stable equilibrium in the rotating reference frame. However, numerical methods often introduce small errors in the calculation of the potential, which can affect the precision of the orbit's behavior, especially near these delicate saddle points (as explained by [38]). As a result, miscalculations may arise if the simulation setup doesn't sufficiently resolve the complex gravitational interactions or if numerical approximations lead to instability in these regions. For the centroid, the Lagrange point of the host gas giant 47 Uma b is at $a = 2.1$, $f = 60°$, and the semi-major axis of each stable regions for each planetary configurations seems to be approximately near the Lagrange point's, while the angular distance of the centroids seem to deviate around $4-7°$, which is relatively near the Lagrange point. This leads to a confirmation of the stability region existing around the Lagrange point of this system.

It can also be inferred that the mass of the host gas giant has effects on the size of the stability region. In the solo host gas giant accompanying the hypothetical Trojan, 47 UMa b has a larger mass than HD 108874

Similar results are present in the HD 10887 system. The fraction of the stable region decreases from $n = 1$ planet to $n = 2$ planets, from 38% to 26%. The Lagrange point of the host gas giant HD 108874 b is at $a = 1.42$,$f = 60°$. There are deviations in the centroid of the stable region from the Lagrange point: the semi-major axis deviates

in around 0.4 AU, and the angular distance deviation is around $6 - 8°$. The centroid coordinates in the stable regions of the MEGNO in this system suggests the final stable region is still around the Lagrange point but not exactly on it. Nonetheless, the existence of the region also leads to a confirmation of the stability region existing near the Lagrange point, as with the 47 UMa system.

Schwarz et al. [8] identified stable regions for Trojan planets in single-planet systems using Lyapunov Characteristic Indicators (LCIs), with stability centered around the Lagrange points for a range of semi-major axes and eccentricities. This study's MEGNO-based stability maps for multi-planet systems show consistent stability regions around the L_4 points, similar to the LCI maps produced by Schwarz et al. However, the presence of additional planets introduces distortions in these regions, particularly near planetary resonances, which were not explored in Schwarz's single-planet models. Additionally, unlike Sudol and Haghighipour [17], who analyzed Trojan stability in circumbinary planetary systems, this study focused on the dynamics of Earth-mass Trojans in multi-planet systems, providing insights into the effects of planetary multiplicity on stability. These results suggest that the gravitational interactions in multi-planet systems significantly alter Trojan stability compared to single-planet models, underscoring the importance of considering planetary multiplicity in future studies of habitable zone Trojans. Overall, this study complements and extends the findings of prior work by providing the first MEGNO-based analysis of Earth-mass Trojan stability in multi-planet systems, highlighting new challenges and opportunities for identifying stable configurations.

While the results demonstrate significant stability regions for hypothetical Trojans, the scope of this study is limited to short-term dynamics and neglects additional physical effects. This work focuses on short-term stability of hypothetical Trojans, limited to 1000 orbits of each simulation. This does not fully take into account for the long-term evolution of Trojan dynamics, even if this timeframe is sufficient to capture initial stability trends. A simple test for long-term dynamics is done, using the 47 UMa system with two planets, integrated for the following timescales: 10^3, 10^4, 10^5, and 10^6 years, all with equal timesteps.

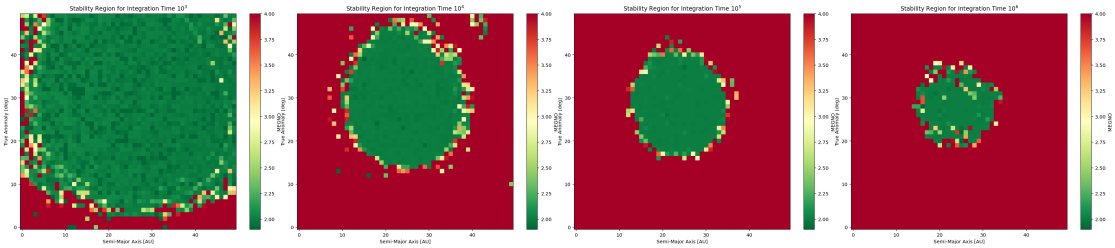

Figure 15: MEGNO heatmaps of the 47 UMa system with the hypothetical Trojan planet simulated for different timescales.

From the Figure 15, longer timescales significantly shrink the stability region size, as previously stated by Schwarz et al. [8]. Here, perturbations from other planetary bodies, slow eccentricity growth, or chaotic dynamics near the boundaries of stable regions could influence Trojan stability [7, 8, 17, 24]. Incorporating longer integrations in future studies would provide a deeper understanding of these trends. Additionally, physical effects such as tidal forces, relativistic corrections, and magnetic interactions, though neglected in this study for computational efficiency, could influence stability in scenarios involving close encounters or highly eccentric orbits. These limitations should be considered when extrapolating the results to long-term scenarios.

Additionaly, only two systems (47 UMa and HD 108874) are simulated in detail due to computational limitations. It is worth noting that the small sample size reduces the generalizability of the conclusions. Considering there are ten unused planetary system parameters in this study, it is worth discussing them so that future work could use them as foundations of studies in astrodynamics.

In the context of habitability, while this study focuses on the orbital stability of Earth-mass Trojans in multi-planet systems, it raises intriguing implications for habitability. Stable orbits near the Lagrange points in the habitable zones of 47 UMa and HD 108874 suggest that these regions could host environments conducive to retaining atmospheres and moderate surface conditions. However, this analysis does not account for critical factors such as stellar radiation, tidal heating, or the potential effects of gas giant magnetic fields, which could significantly impact habitability as discussed by other works [52, 18]. Stable regions identified in this study represent a starting point for future research that incorporates atmospheric modeling, energy budget analysis, and radiative effects.

Thus, the simulations have provided key insights into hypothetical Trojans in multi-planetary systems 47 UMa and HD 108874. Despite having nearby planetary systems, Earth-mass Trojan planets can exist inside of a region surrounding the Lagrange points. While these simulations provide valuable data, the limited timescale and initial parameter grid warrant further exploration to confirm the long-term dynamical trends. Future studies could extend these simulations to incorporate additional planetary systems and refine the parameter space to further elucidate the stability dynamics.

Chapter 5

CONCLUSION AND RECOMMENDATION

5 Conclusion and Recommendation

This study investigated the short-term orbital dynamics and stability of hypothetical Earth-mass Trojan bodies co-orbiting gas giants in the habitable zones (habitable zone) of multi-planetary systems, using the 47 UMa and HD 108874 systems as test cases. Through numerical simulations leveraging the REBOUND N-body code, the study analyzed how variations in initial orbital parameters, such as semi-major axis and true anomaly, and the number of planets, impact the stability of these Trojans.

Key findings indicate that stability regions for Trojans are heavily influenced by the mass of the host gas giant, the initial angular distance from the Lagrange points, and by the gravitational interactions with other planets in the system. Comparison of analysis to previous studies has shown that the stability region around Lagrange points is still measurable in both single and multi-planetary systems. MEGNO analysis further demonstrated the sensitivity of orbital stability to initial conditions, with symmetric patterns in parameter space suggesting periodic or quasi-periodic behavior. Final eccentricity and angular distance analysis suggest that multi-planet systems can help stabilize hypothetical Trojans in these systems.

This study used short-term timescales in studying the stability regions of the hypothetical Trojan planets. This is mainly due to the computational nature of the study, and is limited by software and hardware availabiilities. Despite this, a test for long-term integrations was also done to compare the effects of long-term integrations. It has shown that long-term integrations significantly reduce the size of the stability region.

This research highlights the importance of multi-planet system architecture in deter-

mining Trojan stability, particularly for systems with gas giants in the habitable zone. While previous studies often focused on single-planet systems, this work underscores the complexity introduced by additional planets and their gravitational influences. The results also suggest that stable Trojans in such systems could be potential candidates for future searches for habitable worlds. If stable regions for Trojan planets exist within the habitable zone of a star system, they could also harbor smaller planets or moons, potentially expanding the search for habitable worlds and possible planetary architectures.

Future research could expand on these findings by incorporating additional physical effects, such as tidal forces or relativistic corrections, and exploring a wider range of orbital parameters. Extending the integration timescales could also provide deeper insights into long-term stability and potential habitability. There are also unused planetary systems that fit the criteria during the filtering process, and as such could be the basis of future astronomical research.

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Appendix

Relative energy error calculation for the WHFast integrator: As an additional note, integrators such as WHFast accumulate errors over time, with the rate of accumulation depending on several factors such as timestep, the total simulation time, and the complexity of the system. As a test, the REBOUND environment can calculate for the relative energy error that arises in the timestep used. Here the outer Solar System is used as a test for the energy errors. Studies shave shown that the system is stable and as such is applicable for the test. As shown in Figure 16, timesteps of 10^{-2} provide errors up to 10^{-7}, and a timestep clorser to 1 provides larger errors. Other integrators have shown lower errors, however this test provides enough evidence for the applicability of the WHFast integrator.

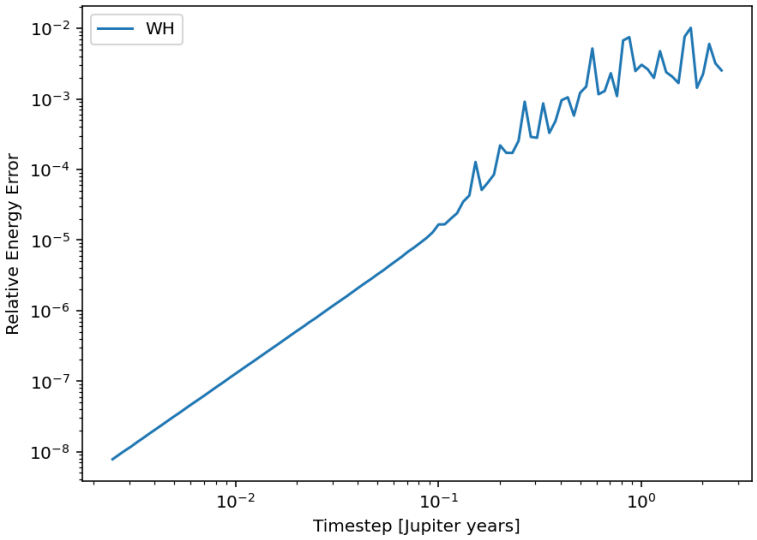

Figure 16: Relative energy error plot for different timesteps using the WHFast integrator applied to the outer Solar System planets.

Code for measuring the relative energy error calculation for the WHFast integrator:

```
import rebound
import numpy as np
import matplotlib.pyplot as plt

from rebound import data
sim = rebound.Simulation()
data.add_outer_solar_system(sim) # either this, or add
    the planets manually
rebound.OrbitPlot(sim)
```

```
# We'll integrate the outer Solar System for 1000 years
    into the future and measure
# the energy error along the way. We do that at random
    intervals to make sure we
# don't have any aliasing with an orbital period. The
    following function runs the
# simulation, and then returns the error.

def measure_energy(sim):
    Nsamples = 1000
    tmax = 2.*np.pi*1e3 # 1000 years
    t_samples = tmax*np.sort(np.random.random(Nsamples))
    E0 = sim.energy() # initial energy
    Emax = 0. # maximum energy error
    for t in t_samples:
        # we do not want to change the timestep to reach
            t exactly, thus
        # we need to set exact_finish_time=False and
            slighlty overshoot.
        sim.integrate(t,exact_finish_time=False)
        E = sim.energy()
        Emax = max(Emax, np.abs((E-E0)/E0))
    return Emax

# To get an idea how our integrators are behaving, we
    want to run this simulation
# for various timesteps. So let us set up an array of
    timesteps from 0.001 to 1
# orbital periods of Jupiter.

N_dt_samples = 100
dt_samples = sim.particles[2].P *
    np.logspace(-3,0.,N_dt_samples)

# Let's run the simulations with the standard WH
    integrator first. We set safe_mode
# to 0 to speed up the calculation.

Emax_wh = np.zeros(N_dt_samples)
for i, dt in enumerate(dt_samples):
    sim_run = sim.copy() # make a copy of the simulation
        so we don't need to set a new one up every time
    sim_run.integrator = "whfast"
    sim_run.dt = dt
```

```
    sim.ri_whfast.safe_mode = False
    Emax_wh[i] = measure_energy(sim_run)

f,ax = plt.subplots(1,1,figsize=(7,5))
ax.set_xscale("log")
ax.set_yscale("log")
ax.set_xlabel("Timestep [Jupiter years]")
ax.set_ylabel("Relative Energy Error")
ax.plot(dt_samples/sim.particles[1].P,Emax_wh,label="WH")
ax.legend();
```

Code for the simulation of each planetary system: (The same code was used for every exoplanetary system with the values changed using the available data from the exoplanet parameter sheet.)

```
import numpy as np
import rebound

# Function to calculate MEGNO for a given semi-major
   axis and true anomaly
def calculate_megno(a, f, planets):
    """
    Calculate MEGNO and final orbital parameters for a
       Trojan in a given planetary system.

    Args:
        a (float): Semi-major axis of the Trojan.
        f (float): True anomaly of the Trojan (radians).
        planets (list): List of dictionaries defining
           the planetary system.

    Returns:
        tuple: MEGNO, final eccentricity, and final
           angular distance (in degrees).
    """
    sim = rebound.Simulation()
    sim.units = ('AU', 'yrs', 'Msun')
    sim.integrator = "whfast"
    sim.ri_whfast.safe_mode = 0

    # Add central star
    sim.add(m=1.06)
```

```
    # Add the Trojan body
    sim.add(m=3e-6, a=a, e=0.032, inc=0, f=f)

    # Add planets and Trojan host
    for planet in planets:
        sim.add(**planet)

    # Activate MEGNO and configure simulation
    sim.move_to_com()
    sim.init_megno()
    sim.exit_max_distance = 20.0
    orbper = 1078 / 365
    sim.dt = 0.01 * orbper
    total_time = 1000 * orbper

    try:
        sim.integrate(total_time, exact_finish_time=0)
        megno = sim.megno()
        final_e = sim.particles[1].e
        final_theta =
           abs(np.rad2deg(sim.particles[1].theta -
           sim.particles[2].theta))
        return megno, final_e, final_theta
    except rebound.Escape:
        final_e = sim.particles[1].e
        final_theta =
           abs(np.rad2deg(sim.particles[1].theta -
           sim.particles[2].theta))
        return 10.0, final_e, final_theta


# Function to generate MEGNO maps for a system
def generate_megno_maps(grid, a_host, planets,
   output_file, header):
    """
    Generate MEGNO maps for a specified planetary system.

    Args:
        grid (int): Resolution of the grid (number of
           values for semi-major axis and true anomaly).
        a_host (float): Semi-major axis of the Trojan
           host planet.
        planets (list): List of dictionaries defining
           the planetary system.
```

```
        output_file (str): Output filename for the CSV.
        header (str): Header for the CSV file.
    """
    # Grid settings
    a_min = a_host - 0.05 * a_host
    a_max = a_host + 0.05 * a_host
    a_values = np.linspace(a_min, a_max, grid)
    f_values_rad = np.linspace(np.deg2rad(30),
       np.deg2rad(90), grid)

    # Initialize storage for results
    megno_list = []

    # Calculate MEGNO for each combination of semi-major
       axis and true anomaly
    for a in a_values:
        for f in f_values_rad:
            megno_value, final_e, final_theta =
               calculate_megno(a, f, planets)
            megno_list.append([a, f, megno_value,
               final_e, final_theta])

    # Save results to a CSV file
    megno_array = np.array(megno_list)
    np.savetxt(output_file, megno_array, delimiter=",",
       header=header, comments="")


# Define planetary systems
systems = [
    {
        "name": "47 UM 1planet",
        "a_host": 2.1,
        "planets": [{"m": 0.002415138, "a": 2.1, "e":
           0.032, "inc": 0}],
        "output_file": "47 UM 1planet.csv",
        "header": "a1,f1,megno1,final_e1,final_theta1",
    },
    {
        "name": "47 UM 2planet",
        "a_host": 2.1,
        "planets": [
            {"m": 0.002415138, "a": 2.1, "e": 0.032,
               "inc": 0},
```

```
            {"m": 0.0005154777, "a": 3.6, "e": 0.098,
                "inc": 0},
        ],
        "output_file": "47 UM 2planet.csv",
        "header": "a2,f2,megno2,final_e2,final_theta2",
    },
    {
        "name": "47 UM 3planet",
        "a_host": 2.1,
        "planets": [
            {"m": 0.002415138, "a": 2.1, "e": 0.032,
                "inc": 0},
            {"m": 0.0005154777, "a": 3.6, "e": 0.098,
                "inc": 0},
            {"m": 0.001565525, "a": 11.6, "e": 0.16,
                "inc": 0},
        ],
        "output_file": "47 UM 3planet.csv",
        "header": "a3,f3,megno3,final_e3,final_theta3",
    },
]

# Generate MEGNO maps for all systems
grid = 50
for system in systems:
    print(f"Processing {system['name']}...")
    generate_megno_maps(
        grid=grid,
        a_host=system["a_host"],
        planets=system["planets"],
        output_file=system["output_file"],
        header=system["header"],
    )
```

Code for the plotting of heatmaps from generated data:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Read and process orbit data
df1 = pd.read_csv('47 UM 1planet.csv')
df2 = pd.read_csv('47 UM 2planet.csv')
```

```
df3 = pd.read_csv('47 UM 3planet.csv')
orbit_data = pd.concat([df1, df2, df3], axis=1)

# Convert angular data to degrees
f_deg = np.rad2deg(orbit_data[['f1', 'f2', 'f3']])

# Axis limits
a_min, a_max = orbit_data['a1'].min(),
    orbit_data['a1'].max()
f_min, f_max = f_deg['f1'].min(), f_deg['f1'].max()

# Common plotting functions
def plot_subplot(ax, pivot_data, cmap, vmin, vmax,
    title, xlabel, ylabel, cbar_label, colorbar=True):
    cax = ax.imshow(pivot_data, vmin=vmin, vmax=vmax,
        cmap=cmap, origin="lower", aspect='auto',
                    extent=[a_min, a_max, f_min, f_max])
    ax.scatter(2.1, 60, color='blue', s=100, marker="x")
    ax.text(2.11, 60, '$L_4$', color='white',
        fontsize=12)
    ax.set_xlabel(xlabel)
    ax.set_ylabel(ylabel)
    ax.set_title(title)
    ax.set_facecolor('gray')
    if colorbar:
        fig.colorbar(cax, ax=ax, label=cbar_label)

# MEGNO Plots ==========================
fig, axs = plt.subplots(1, 3, figsize=(18, 6),
    gridspec_kw={'width_ratios': [1, 1, 1.2]})
cmap_megno = "RdYlGn_r"

for i, (f_col, a_col, megno_col, title) in
    enumerate(zip(['f1', 'f2', 'f3'], ['a1', 'a2', 'a3'],
    ['megno1', 'megno2', 'megno3'], ['47 UMa b', '47 UMa
    b, c', '47 UMa b, c, d'])):
    pivot_data = orbit_data.pivot(index=f_col,
        columns=a_col, values=megno_col)
    plot_subplot(axs[i], pivot_data, cmap_megno,
        vmin=1.9, vmax=4, title=title,
                xlabel="Semi-Major Axis [AU]",
                    ylabel="True Anomaly [$\theta$]",
                    cbar_label='MEGNO', colorbar=(i == 2))
```

```
plt.tight_layout()
plt.savefig("47 UMa System - MEGNO.png",
    bbox_inches='tight', pad_inches=0)
plt.show()

# Eccentricity Plots ========================
fig, axs = plt.subplots(1, 3, figsize=(18, 6),
    gridspec_kw={'width_ratios': [1, 1, 1.2]})
cmap_stable = plt.get_cmap('viridis')
cmap_stable.set_under('gray')

# Process eccentricity
for i, (megno_col, ecc_col) in enumerate(zip(['megno1',
    'megno2', 'megno3'], ['final_e1', 'final_e2',
    'final_e3'])):
    orbit_data.loc[(orbit_data[megno_col] <= 1.9) |
        (orbit_data[megno_col] >= 2.1), ecc_col] = -1

for i, (f_col, a_col, ecc_col, title) in
    enumerate(zip(['f1', 'f2', 'f3'], ['a1', 'a2', 'a3'],
    ['final_e1', 'final_e2', 'final_e3'], ['47 UMa b', '47
    UMa b, c', '47 UMa b, c, d'])):
    pivot_data = orbit_data.pivot(index=f_col,
        columns=a_col, values=ecc_col)
    plot_subplot(axs[i], pivot_data, cmap_stable,
        vmin=0, vmax=0.5, title=title,
                xlabel="Semi-Major Axis [AU]",
                    ylabel="True Anomaly [$\theta$]",
                    cbar_label='Eccentricity',
                    colorbar=(i == 2))

plt.tight_layout()
plt.savefig("47 UMa System - Final Eccentricity.png",
    bbox_inches='tight', pad_inches=0)
plt.show()

# Angular Distance Plots ======================
fig, axs = plt.subplots(1, 3, figsize=(18, 6),
    gridspec_kw={'width_ratios': [1, 1, 1.2]})

# Process angular distance
for i, (megno_col, theta_col) in
    enumerate(zip(['megno1', 'megno2', 'megno3'],
    ['final_theta1', 'final_theta2', 'final_theta3'])):
```

```
    orbit_data.loc[(orbit_data[megno_col] <= 1.9) |
        (orbit_data[megno_col] >= 2.1), theta_col] = -1

for i, (f_col, a_col, theta_col, title) in
    enumerate(zip(['f1', 'f2', 'f3'], ['a1', 'a2', 'a3'],
    ['final_theta1', 'final_theta2', 'final_theta3'], ['47
    UMa b', '47 UMa b, c', '47 UMa b, c, d'])):
    pivot_data = orbit_data.pivot(index=f_col,
        columns=a_col, values=theta_col)
    plot_subplot(axs[i], pivot_data, cmap_stable,
        vmin=0, vmax=None, title=title,
                    xlabel="Semi-Major Axis [AU]",
                        ylabel="True Anomaly [$\theta$]",
                        cbar_label='Angular Distance
                        [$\theta$]', colorbar=(i == 2))

plt.tight_layout()
plt.savefig("47 UMa System - Final Angular
    Distance.png", bbox_inches='tight', pad_inches=0)
plt.show()
```

Code for the plotting of heatmaps from generated data:

```
import numpy as np
import pandas as pd

# Read the data from CSV files
data1 = pd.read_csv('../47 UMa/47 UM total.csv')
data2 = pd.read_csv('../HD 108874/HD 108874 total.csv')

# Function to calculate the stable fraction
def calculate_stable_fraction(data, column_name):
    stable_count =
        data[column_name].value_counts().get(1, 0)  #
        Count of 1s (stable)
    total_count = len(data[column_name])  # Total entries
    if total_count > 0:
        return stable_count / total_count
    return 0  # Avoid division by zero

# Function to calculate the centroid of the stable region
def centroid_stable_region(data, a_column, f_column,
    stable_column):
```

```
    a = data[a_column]
    f = np.rad2deg(data[f_column])
    stability = data[stable_column]

    weighted_sum_a = np.sum(a * stability)  # Weighted
       sum of semi-major axis
    weighted_sum_f = np.sum(f * stability)  # Weighted
       sum of true anomaly
    total_weight = np.sum(stability)  # Total weight
       (sum of stability values)

    if total_weight > 0:
        centroid_a = weighted_sum_a / total_weight  #
           Centroid of a
        centroid_f = weighted_sum_f / total_weight  #
           Centroid of f
    else:
        centroid_a = centroid_f = 0  # Avoid division by
           zero

    return centroid_a, centroid_f  # Return the centroids

# Calculate stable fractions for each table
stable_fraction_1 = calculate_stable_fraction(data1,
   'stable1')  # 47 UMa - 1 planet
stable_fraction_2 = calculate_stable_fraction(data1,
   'stable2')  # 47 UMa - 2 planets
stable_fraction_3 = calculate_stable_fraction(data1,
   'stable3')  # 47 UMa - 3 planets
stable_fraction_4 = calculate_stable_fraction(data2,
   'stable1')  # HD 108874 - 1 planet
stable_fraction_5 = calculate_stable_fraction(data2,
   'stable2')  # HD 108874 - 2 planets

# Calculate centroid point
centroid_point_1 = centroid_stable_region(data1, 'a1',
   'f1', 'stable1')  # 47 UMa - 1 planet
centroid_point_2 = centroid_stable_region(data1, 'a2',
   'f2', 'stable2')  # 47 UMa - 2 planets
centroid_point_3 = centroid_stable_region(data1, 'a3',
   'f3', 'stable3')  # 47 UMa - 3 planets
centroid_point_4 = centroid_stable_region(data2, 'a1',
   'f1','stable1')  # HD 108874 - 1 planet
centroid_point_5 = centroid_stable_region(data2, 'a2',
```

```
    'f2','stable2')  # HD 108874 - 2 planets

# Create a summary table with the calculated stable
    fractions
summary_table = pd.DataFrame({
    'Planet configurations': [
        '47 UMa - 1 planet',
        '47 UMa - 2 planets',
        '47 UMa - 3 planets',
        'HD 108874 - 1 planet',
        'HD 108874 - 2 planets'
    ],
    'Stable Fraction': [
        stable_fraction_1 ,
        stable_fraction_2 ,
        stable_fraction_3 ,
        stable_fraction_4 ,
        stable_fraction_5
    ],
    'Centroid Point': [
        centroid_point_1 ,
        centroid_point_2 ,
        centroid_point_3 ,
        centroid_point_4 ,
        centroid_point_5 ,
    ]
})

# Display the summary table
summary_table.to_csv('summary_table.csv', index=False)
```

Stellar parameters of the remaining 10 unused systems that fit the criteria during the filtering process:

System	Mass $[M_\odot]$	Radius $[R_\odot]$	Effective Temperature [K]	Luminosity $[L_\odot]$	Inner habitable zone [AU]	Outer habitable zone [AU]
HD 13908	1.29	1.67	6255	3.846235542	1.471912732	3.478345153
HD 159868	1.19	2.13	5534	3.833623905	1.469497582	3.4726378
HD 1605	1.33	3.49	4915	6.403812697	1.89925568	4.488219068
HD 163607	1.12	1.76	5522	2.594804385	1.208972536	2.856978997
HD 45364	0.82	0.89	5466	0.637018729	0.599018484	1.415568326
HD 82943	1.08	1.18	5935	1.556469397	0.936341811	2.212712703
KELT-6	1.13	1.53	6272	3.263628663	1.355858889	3.204092943
Kepler-419	1.39	1.74	6430	4.662684893	1.620623592	3.829770675
TOI-1669	0.99	1.06	5497	0.924290218	0.721553108	1.705135571
TOI-4600	0.89	0.81	5170	0.422304674	0.487727303	1.152570979

Planetary parameters of the remaining 10 unused systems:

Planet	Mass $[M_J]$	a [AU]	e	inc [deg]
HD 13908 b	0.865	0.514	0.02	0
HD 13908 c*	5.13	2.03	0.12	0
HD 159868 b*	2.218	2.32	0.024	0
HD 159868 c	0.768	1.032	0.184	0
HD 1605 b	0.934	1.492	0.095	0
HD 1605 c*	3.62	3.584	0.099	0
HD 163607 b	0.7836	0.362	0.7441	0
HD 163607 c*	2.201	2.39	0.08	0
HD 45364 b	0.1893	0.6793	0.067	0
HD 45364 c*	0.549	0.902	0.019	0
HD 82943 b*	1.681	1.18306	0.162	90
HD 82943 c	1.959	0.74345	0.3663	90
KELT-6 b	0.52	1.18306	0.22	88.81
KELT-6 c*	3.71	2.39	0.21	0
Kepler-419 b	2.5	0.37	0.833	88.95
Kepler-419 c*	7.3	1.68	0.184	88
TOI-1669 b*	0.60095	1.27	0	0
TOI-1669 c	0.0409	0.0376	0	0
TOI-4600 b	3.02	0.349	0.25	89.76
TOI-4600 c*	9.27	1.152	0.21	89.9